AGENTIC AI FOR ROBUST AND SOCIALLY AWARE AUTONOMOUS DRIVING

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Autonomous vehicles operating at unsignalized intersections must coordinate right-of-way while dealing with uncertain maneuver intentions, conflicting paths, traffic rules, and limited local information. This research presents Multi-Agent Negotiation for Right-of-Way in Complex Intersections, a decentralized framework that combines ego-local perception, movement-intention estimation, spatial and temporal conflict reasoning, explicit regulatory precedence, constrained Multi-Agent Proximal Policy Optimization (MAPPO), and vehicle-owned physical execution. Real-world trajectories from the inD dataset were used to develop GRU-based intention models, while the complete right-of-way framework was implemented and evaluated in SUMO. The deployed intention models achieved 89.66% test accuracy at a 1.0 s prediction horizon and 93.10% at 0.5 s. Three predefined MAPPO configurations were compared, and the five-epoch E5 configuration was selected using validation Total Team Travel Time. Targeted validation included rule-resolved, negotiation-required, complex turning, dense continuous-traffic, and communication-fault scenarios. The rule-resolved two-vehicle case and the three-vehicle multiple-left-turn case completed without observed collisions or blocked-zone violations, and runtime integrity checks confirmed decentralized execution without centralized critic calls. However, dense 16-vehicle validation exposed a major liveness limitation: only one vehicle completed in the nominal dense case, despite the absence of observed collision or blocked-zone violations. Analysis indicated that inconsistent ego-local intention and precedence beliefs could create conservative holding and deadlock. The research therefore demonstrates bounded simulation-level feasibility for constrained decentralized right-of-way negotiation while identifying decentralized belief reconciliation and dense-traffic liveness as key areas for future work.

Keywords: Autonomous vehicles, Multi-agent reinforcement learning, MAPPO, Right-of-way negotiation, Unsignalized intersections

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AIM	Autonomous Intersection Management
AV	Autonomous Vehicle
CAV	Connected Autonomous Vehicle
CTDE	Centralized Training with Decentralized Execution
DRL	Deep Reinforcement Learning
FCFS	First-Come First-Served
GNN	Graph Neural Network
GRU	Gated Recurrent Unit
LDM	Local Dynamic Map
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MPNN	Message Passing Neural Network
ODD	Operational Design Domain
ONNX	Open Neural Network Exchange
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
SOTIF	Safety of the Intended Functionality
StVO	Straßenverkehrs-Ordnung
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
TTT	Total Team Travel Time
V2V	Vehicle-to-Vehicle

1 INTRODUCTION

1.1 Overall project context

Autonomous driving systems are becoming increasingly capable of handling routine driving tasks such as lane following, obstacle detection, speed control, and path planning. However, urban driving still presents situations in which a vehicle must do more than simply react to visible objects. It must interpret uncertain behavior, anticipate what other road users may do next, coordinate with other vehicles, and make decisions that are both safe and socially acceptable. These challenges become particularly important in locations such as intersections, pedestrian crossings, merging areas, and partially occluded urban roads.

The broader research project, **Agentic AI for Robust and Socially Aware Autonomous Driving**, investigates this problem through four complementary research components. Rather than treating autonomous driving only as a perception or control problem, the project considers how intelligent vehicles can reason about hidden intentions, coordinate with other agents, behave appropriately around people, and learn from experiences shared across a fleet.

The first component, **Adaptive Intent Prediction in Occluded Urban Scenarios**, investigates situations in which pedestrians or other vulnerable road users are only partially visible. Its focus is to use partial observations and contextual evidence to predict possible intent before the road user becomes fully visible. This addresses an important limitation of purely reactive perception systems, which may recognize a dangerous situation only after the road user has already entered the vehicle's visible area.

The second component, and the focus of this thesis, is **Multi-Agent Negotiation for Right-of-Way in Complex Intersections**. It addresses interactions among autonomous vehicles at unsignalized intersections, where there is no traffic signal or central traffic controller explicitly telling every vehicle when to move. The component studies how autonomous vehicles can reason right-of-way, identify

conflicts, exchange coordination information, and negotiate when a clear ordering cannot be obtained from mandatory traffic rules alone.

The third component, **Proactive Social Compliance Modeling using Agentic Theory of Mind**, focuses on interactions between autonomous vehicles and pedestrians. Instead of considering only a pedestrian's physical trajectory, the component attempts to infer behaviorally meaningful states such as hesitation, commitment, distraction, or risky behavior and use these states to generate more socially appropriate driving decisions.

The fourth component, **Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing**, considers learning at fleet level. Its purpose is to allow a vehicle that encounters a rare or useful road situation to determine whether that knowledge should be shared with other vehicles, while filtering unreliable information and avoiding unnecessary sharing of raw data.

Together, these components address different stages of socially aware autonomous driving: anticipating hidden behavior, coordinating with other vehicles, interpreting social behavior, and learning collectively across vehicles. They are complementary research problems rather than four implementations of the same algorithm. This thesis concentrates only on the second problem: how autonomous vehicles can coordinate right-of-way at complex unsignalized intersections.

This problem remains relevant because intersections continue to be difficult environments for automated driving. A recent review of decision-making at unsignalized intersections describes them as highly uncertain multi-agent environments and concludes that a robust approach suitable for real-world use has not yet been established [1]. Recent crash-sequence research also found that intersections occurred much more frequently in the analyzed automated-vehicle crashes than in comparable human-driven-vehicle crashes, with intersection involvement reported as 68.5% and 34.9%, respectively, in the studied datasets [2]. These results should not be interpreted as a universal crash-rate comparison, but they reinforce the need for improved decision-making and coordination in intersection scenarios.

1.2 Individual component context

The individual research presented in this thesis is titled **Multi-Agent Negotiation for Right-of-Way in Complex Intersections**. The work focuses on an unsignalized four-way intersection used by autonomous vehicles approaching from the north, east, south, and west. A vehicle may continue straight, turn left, or turn right, producing twelve legal movement combinations within the selected intersection model.

At first sight, deciding who should move through an intersection may appear to be a simple matter of applying traffic rules. In practice, however, the problem becomes more difficult when several vehicles arrive within a short period of one another. Their trajectories may cross or merge, their intended maneuvers may initially be uncertain, and each vehicle may observe the situation from a different position. Some movements can safely occur at the same time, while others require one vehicle to yield.

The difficulty is therefore not simply detecting another vehicle. An autonomous vehicle needs to determine whether that vehicle is relevant to its own planned route, what movement the other vehicle is likely to perform, whether the two movement paths share a physical conflict area, whether both vehicles could reach that area at similar times, and what the applicable traffic rules require.

The final system developed in this research uses **SUMO** as the microscopic traffic simulator and **TraCI** as the interface between the simulation and the research implementation. Each autonomous vehicle is treated as an individual decision-making agent. Instead of using a perfect global state as its runtime input, each vehicle maintains an ego-local view of the surrounding environment through a Local Dynamic Map. Recent motion information is also used by a learned intention-prediction model to estimate whether relevant vehicles are likely to travel straight, turn left, or turn right.

These local observations are then combined with map-derived movement paths, spatial conflict zones, temporal occupancy reasoning, and explicit traffic-rule precedence. The purpose of these deterministic stages is to identify relationships that should not be decided by reinforcement learning. If the traffic rules already require one vehicle to yield to another, the learned policy is not allowed to override that obligation.

Only relationships that remain legitimately negotiation-eligible are passed to a learned negotiation layer based on **Multi-Agent Proximal Policy Optimization (MAPPO)**. MAPPO belongs to the family of cooperative multi-agent reinforcement-learning methods based on Proximal Policy Optimization and has demonstrated strong performance across cooperative multi-agent tasks [3]. In this research, its role is deliberately limited: it negotiates semantic right-of-way claims rather than directly producing steering, braking, or acceleration commands.

The outcome of negotiation is subsequently translated into physical driving constraints by the execution layer. Each autonomous vehicle independently determines whether it is permitted to proceed toward a relevant conflict zone or whether it should continue to hold before entering it. The physical execution stage therefore remains decentralized, while SUMO's native safety mechanisms remain active as an additional safeguard.

The final architecture is consequently a **hybrid decision-making framework**. Traffic rules and physical conflict constraints are handled explicitly, while reinforcement learning is used only for the part of the coordination problem in which negotiation is permitted. This distinction is central to the final research design.

1.3 Problem statement

Safe traversal of an unsignalized intersection requires autonomous vehicles to make coordinated decisions without the explicit movement phases provided by traffic lights. The problem becomes especially difficult when several vehicles approach the intersection at similar times and follow movements that overlap within the same conflict areas.

A vehicle cannot base its decision only on who arrived first, or which vehicle is physically closest to the intersection. A nearby vehicle may be following a non-conflicting route, while another vehicle approaching from a different direction may be on a path that intersects the ego vehicle's movement. The situation is further complicated by uncertainty about the intended maneuvers of other vehicles and by the fact that each autonomous vehicle observes the environment from its own local perspective.

Rule-based approaches provide an important foundation because right-of-way obligations must be respected. However, fixed ordering strategies alone may become overly restrictive or may fail to resolve some multi-vehicle interactions efficiently. Centralized autonomous intersection-management systems provide another solution by collecting information from vehicles and computing a common decision. Previous research has shown that such approaches can substantially improve intersection efficiency [4]. However, they rely on a central decision mechanism and therefore do not represent the decentralized runtime behavior studied in this research.

Decentralized learning has also been investigated in previous work. DCL-AIM, for example, decomposes intersection management into coordination problems between interacting agents and dynamically identifies where cooperation is required [5]. This demonstrates that decentralized coordination is feasible, but the practical problem remains of combining partial local observations with uncertain intentions, regulatory precedence, physical conflict reasoning, and safe execution.

Learning alone also creates difficulties in a safety-critical environment. Feng et al. found that a multi-agent reinforcement-learning policy did not by itself ensure complete compliance with right-of-way rules and therefore introduced a dedicated

right-of-way cooperator to improve regulatory compliance and safety [6]. This observation is particularly important for the present research because it suggests that mandatory traffic rules should not simply be treated as behaviors that an agent may or may not learn.

Accordingly, the main problem addressed in this thesis is: How can multiple autonomous vehicles coordinate right-of-way at a complex unsignalized intersection using decentralized, ego-local information while preserving mandatory traffic-rule and physical-safety constraints and using learned negotiation only where a genuine coordination choice remains?

A useful solution must also consider liveness and efficiency. Preventing collisions by stopping every vehicle indefinitely would be safe in a very limited sense, but it would not constitute effective intersection management. The system must therefore be examined in terms of whether vehicles clear the intersection, how long they wait, how traffic flow changes under congestion, and whether some approaches or vehicles are disadvantaged by the coordination process.

1.4 Research gap

A considerable body of research already exists on autonomous intersection management. Therefore, the research gap addressed in this thesis is not simply that reinforcement learning or decentralized control has never been used at intersections.

Earlier autonomous-intersection-management approaches often rely on an intersection manager, reservation mechanism, or other form of centralized coordination. Learning-based approaches have extended this idea by allowing a central or shared decision process to adapt to dynamic traffic. For example, the advanced RAIM system proposed by Guillen-Perez and Cano applies multi-agent deep reinforcement learning to collaborative control at intersections and reports substantial reductions in travel and congestion-related time [4]. Such work demonstrates the potential of learned coordination, but the control structure differs from an architecture in which every vehicle determines its own final execution decision.

Decentralized approaches address part of this limitation. Wu et al. proposed DCL-AIM, in which the multi-agent problem is decomposed, and coordination is concentrated on states where interaction is necessary [5]. This is closely related to the idea that not all vehicles at an intersection need to participate in the same negotiation at the same time. However, decentralized coordination alone does not solve the questions of uncertain maneuver intention, regulatory precedence, temporal conflict relevance, or how to prevent a learned negotiation from removing a safety-critical ordering.

Other approaches use game theory rather than reinforcement learning. GamePlan, for example, uses an auction-like game-theoretic mechanism in which driving style influences vehicle priority at intersections, roundabouts, and merging situations [7]. This provides a useful model of interactive behavior, particularly in mixed human-autonomous environments, but it relies on a different coordination model from the iterative local precedence negotiation investigated here.

Another relevant issue is intention uncertainty. The path conflict between two vehicles depends on the maneuvers they intend to perform. The **inD dataset** was

developed specifically to provide naturalistic trajectory data from urban intersections and contains more than 13,500 recorded road users across four German intersections [8]. Such data makes it possible to train and evaluate intention-prediction models using realistic motion histories rather than assuming that every vehicle's future movement is always known.

The gap addressed by this thesis is therefore more specific than the gap stated at the early proposal stage. The aim is not to replace all traffic rules with reinforcement learning. Instead, this work investigates how several elements can be combined within one decentralized right-of-way pipeline: ego-local observation, learned movement-intention estimation, map-based conflict geometry, temporal relevance, explicit traffic-rule precedence, structured inter-vehicle claims, and constrained MAPPO negotiation.

The central design principle is that the system first determines **what cannot legitimately be negotiated**. Mandatory regulatory precedence and physical safety constraints are retained outside the learned policy. MAPPO is used only after these constraints have been considered and only when a negotiation-eligible relationship remains. The resulting coordination commitments are then interpreted independently by each vehicle's execution layer rather than by a central runtime authority.

This combination of constrained learned negotiation with decentralized physical execution under partial local information forms the specific research gap investigated in this thesis.

1.5 Aim and objectives

The main aim of **Multi-Agent Negotiation for Right-of-Way in Complex Intersections** is:

To design, implement, and evaluate a decentralized multi-agent framework that enables autonomous vehicles to coordinate right-of-way at an unsignalized four-way intersection by combining ego-local observation, intention estimation, traffic-rule and conflict reasoning, and constrained MAPPO-based negotiation.

To achieve this aim, the research has six specific objectives:

1. **To develop an ego-local representation of intersection traffic** in which each autonomous vehicle maintains its own Local Dynamic Map and reasons from locally available observations rather than depending on a global runtime controller.
2. **To incorporate movement-intention estimation into right-of-way reasoning** so that surrounding vehicles can be associated with likely straight, left-turn, or right-turn movements when assessing potential conflicts.
3. **To model spatial and temporal conflict relationships and mandatory traffic-rule precedence** before negotiation, ensuring that non-negotiable regulatory and physical constraints remain outside the learned policy.
4. **To develop and train a MAPPO-based negotiation mechanism** for right-of-way relationships that remain explicitly eligible for negotiation, while constraining invalid negotiation actions through hard semantic and safety masks.
5. **To implement decentralized physical execution** in which each autonomous vehicle independently converts valid right-of-way commitments into conflict-zone permission and speed constraints.
6. **To evaluate the completed framework in SUMO** using selected rule-resolved, negotiation-required, complex, congested, and failure-oriented scenarios, with particular attention to safety, completion, waiting time, throughput, fairness, deadlock behavior, and decentralized execution.

These objectives reflect the final implemented research rather than the broader proposal-stage plan.

1.6 Research questions or hypotheses

The research is guided by four questions.

RQ1: To what extent can an ego-local decentralized framework establish consistent right-of-way coordination among autonomous vehicles in selected unsignalized-intersection scenarios?

RQ2: Can MAPPO-based negotiation contribute to resolving negotiation-eligible right-of-way relationships while mandatory traffic-rule precedence and physical-safety constraints remain protected outside the learned policy?

RQ3: How does the behavior of the framework change as interactions become more complex or congested, particularly in terms of completion, waiting time, throughput, fairness, safety, and deadlock?

RQ4: What limitations become visible when decentralized right-of-way coordination is evaluated under difficult interaction patterns, incomplete coordination information, or dense continuous traffic?

These questions deliberately avoid if every test scenario will produce successful traffic flow. Identifying conditions under which the system remains safe but loses lively is also relevant research finding.

1.7 Scope limitations and assumptions

The scope of the research was intentionally narrowed during development so that the final system could be studied in a controlled and reproducible environment.

The system-level study considers a single **unsignalized four-way intersection**. Autonomous vehicles may approach from four directions and may perform left-turn, straight, or right-turn maneuvers. This gives twelve legal movement paths within the selected network. The final system-level experiments consider autonomous vehicles only. Human-driven vehicles, pedestrians, cyclists, signal-controlled intersections, T-junctions, roundabouts, and multi-intersection networks are outside the experimental scope of this individual thesis.

The research is implemented and evaluated primarily in **SUMO**. TraCI is used to exchange state and control information between the simulator and the research implementation. The results therefore represent simulation evidence and should not be interpreted as proof of safe deployment on public roads.

Real-world trajectory data is used to support the intention-prediction part of the component. In particular, the implemented intention models are based on the inD intersection dataset. However, the final right-of-way policy itself is not validated through real vehicles or through direct replay of real-world traffic as if those vehicles had followed the proposed negotiation decisions. The distinction between real-world data for intention modelling and simulation-based system validation is maintained throughout the thesis.

The runtime architecture assumes ego-local information and structured communication between autonomous vehicles. This communication is modelled within the simulation; the research does not attempt to reproduce a complete real V2V radio stack, wireless packet-loss process, cryptographic authentication protocol, or production communication infrastructure. Missing and stale coordination information are considered only through controlled fail-safe tests.

The number of vehicles present in the simulation can be greater than the number of vehicles involved in a single negotiation event. In continuous traffic, following vehicles remain under normal car-following behavior until they become physically

relevant queue-head vehicles. Because the selected intersection has four approaches, the learned negotiation context is limited to a maximum of four simultaneous queue-head participants. This allows congested traffic to be represented without changing the trained policy into an untested large-agent negotiation problem.

The study also separates safety from liveness. Zero observed collisions in a finite group of simulations cannot establish a zero real-world collision probability. Similarly, a vehicle that remains stopped can avoid a collision while still causing an unacceptable deadlock. For this reason, the evaluation considers completion, deadlock, waiting time, throughput, fairness, and safe degradation in addition to collision-related measures.

Finally, the work does not claim that the selected MAPPO policy is a globally optimal solution for every intersection or traffic population. The thesis evaluates the behavior of a bounded research prototype under the defined operational assumptions.

1.8 Individual contribution

The individual contribution of this thesis is the design, development, integration, and evaluation of the research component titled **Multi-Agent Negotiation for Right-of-Way in Complex Intersections** within the wider project **Agentic AI for Robust and Socially-Aware Autonomous Driving**.

My work focuses on the autonomous-vehicle-to-autonomous-vehicle coordination problem at an unsignalized intersection. It includes the construction of the four-way SUMO research environment and its legal movement paths; ego-local vehicle observation and Local Dynamic Map integration; movement-intention inference; representation of movement paths and conflict zones; temporal conflict assessment; traffic-rule precedence reasoning; construction and exchange of local precedence claims; graph-based representation of negotiation contexts; MAPPO-based proposer and responder decision-making; and the conversion of accepted coordination commitments into ego-local physical execution constraints.

A significant part of the contribution is the separation between deterministic and learned decision-making. The learned policy is not responsible for deciding every aspect of vehicle motion. Mandatory right-of-way relationships and physical conflict constraints are established before negotiation, while MAPPO is restricted to cases that the system identifies as legitimately negotiable. This makes it possible to distinguish a rule-resolved decision from a learned negotiation decision and to prevent the reinforcement-learning policy from intentionally overriding mandatory precedence.

The implementation also includes decentralized physical execution. After negotiation, there is no central runtime controller issuing a global instruction such as "vehicle A goes, and vehicle B stops." Instead, each ego vehicle receives the coordination information relevant to itself and independently derives whether it should remain held before a conflict zone or whether it has sufficient permission to proceed.

The final contribution also covers experimental validation and failure analysis. The component was tested using basic rule-resolved cases, negotiation-required cases,

complex turning interactions, continuous four-approach traffic, fairness and waiting-time measurements, communication-fault checks, and deliberately retained difficult cases that expose liveness limitations. The purpose of this evaluation is not to hide unsuccessful cases, but to determine both where the proposed concept operates as intended and where further research is required.

1.9 Thesis structure

The remainder of the thesis follows the structure specified in the individual thesis template. Individual Thesis Content

Chapter 2, Literature Review and Related Work, reviews the theoretical and technical background relevant to autonomous decision-making at unsignalized intersections. It examines rule-based and reservation approaches, centralized and decentralized intersection management, multi-agent reinforcement learning, MAPPO, game-theoretic negotiation, intention prediction, and related autonomous-driving systems. The chapter concludes by refining the research gap addressed by this study.

Chapter 3, Research Methodology, explains the research design, study materials and datasets, SUMO experimental environment, intention-model development, MAPPO training and selection procedure, scenario design, evaluation measures, data analysis methods, ethical considerations, and reproducibility practices.

Chapter 4, Product and Solution Design, presents the final system architecture and explains the design decisions behind the ego-local perception layer, Local Dynamic Map, intention prediction, map and conflict-zone representation, temporal conflict reasoning, regulatory precedence, negotiation graphs, MAPPO negotiation actors, communication protocol, and decentralized execution layer.

Chapter 5, Implementation, Integration and Testing, describes the development environment and technology stack, implementation of the major modules, integration with the broader autonomous-driving project, software testing, regression testing, simulation operation, and demonstration infrastructure.

Chapter 6, Results and Validation, presents the experimental evidence obtained from the completed system. It reports research and technical results, selected complex scenarios, congestion behavior, safety and liveness measurements, fairness and waiting-time evidence, controlled fault behavior, and relevant integration results.

Chapter 7, Discussion, interprets these findings in relation to the research questions and previous literature. Attention is given to the role of constrained negotiation, the

effect of partial local information, safe degradation, fairness, dense-traffic liveness limitations, and threats to validity.

Chapter 8, Commercialization and Sustainability, considers possible users and applications of the proposed approach, commercial and deployment considerations, scalability, intellectual-property issues, compliance requirements, and sustainability implications.

Chapter 9, Conclusion and Future Directions, summarizes the main findings, reviews achievement of the research objectives, identifies the individual research and product contributions, and outlines future work required to improve liveness, scalability, communication realism, generalization to other intersection types, and eventual real-world validation.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review examines research relevant to autonomous right-of-way coordination at unsignalized intersections, with particular attention to decentralized decision-making, multi-agent reinforcement learning, traffic-rule reasoning, intention uncertainty, graph-based interaction modelling, and the evaluation of safety, efficiency, and fairness.

A focused narrative review was adopted rather than a formal systematic literature review. The purpose of the review was not to quantify all publications in autonomous intersection management, but to identify the main technical approaches that directly inform the design choices of this research. Peer-reviewed journal and conference publications were prioritized, while authoritative dataset and foundational algorithm papers were included where they were necessary to explain the technical basis of the implemented framework.

The literature search covered work published primarily between 2019 and 2026, while several earlier foundational studies were retained because they established important ideas such as reservation-based autonomous intersection management and decentralized multi-agent coordination. Sources were identified using combinations of terms including *autonomous intersection management*, *unsignalized intersection*, *connected autonomous vehicles*, *multi-agent reinforcement learning*, *MAPPO*, *right-of-way*, *decentralized coordination*, *vehicle intention prediction*, *conflict graph*, *fairness*, and *intersection negotiation*. IEEE Xplore, ScienceDirect, SpringerLink, Taylor & Francis, and major machine-learning conference repositories were used as the principal academic sources.

Studies were considered particularly relevant when they addressed one or more of the following themes: vehicle coordination at unsignalized intersections, autonomous or connected vehicle right-of-way, multi-agent or decentralized decision-making, reinforcement-learning-based intersection control, interaction or conflict graphs, vehicle-intention uncertainty, explicit traffic-rule constraints, or safety and fairness

evaluation. Studies focusing only on conventional traffic-signal timing were not treated as central related work unless they introduced a coordination principle relevant to autonomous vehicles. Similarly, pedestrian-only decision-making and highway-only coordination were outside the primary scope of this review.

The review was organized into five related areas: the fundamentals of unsignalized-intersection coordination; classical autonomous intersection management; decentralized and learning-based approaches; intention-aware and graph-based reasoning; and the relationship between safety, efficiency, liveness, and fairness. This structure was used to identify which challenges have already been addressed in the literature and which combination of challenges remains insufficiently covered.

Table 1: Scope of the literature review

Review theme	Main question considered	Examples of relevant literature
Unsignalized-intersection decision-making	Why is right-of-way difficult under uncertainty?	Al-Sharman et al. [1]
Autonomous intersection management	How are crossing permissions or orders assigned?	Dresner and Stone [9], Aoki and Rajkumar [10]
Multi-agent and decentralized learning	Can vehicles learn cooperative policies without a central runtime decision-maker?	Wu et al. [5], Yu et al. [3], Zhuang et al. [11]
Right-of-way constrained learning	How can learned decisions coexist with mandatory precedence?	Feng et al. [6]
Interaction graphs	How can multi-vehicle dependencies be represented compactly?	Zhou et al. [12]
Intention uncertainty	How can likely manoeuvres be inferred before crossing?	Bock et al. [8], Cao et al. [17]
Fairness and liveness	Is collision avoidance sufficient for successful coordination?	Wu et al. [13]

2.2 Domain and theoretical foundations

2.2.1 Decision-Making at Unsignalized Intersections

An unsignalized intersection differs fundamentally from a signal-controlled intersection because no traffic phase explicitly assigns movement permission. Instead, the vehicles approaching the junction must determine whether their intended paths conflict and which vehicle is required to yield.

This creates a multi-agent decision problem. The action taken by one vehicle changes the feasible and safe actions of the others. A vehicle that accelerates toward a conflict point may require another vehicle to delay its entry, while a vehicle that yields can create an opportunity for another movement to proceed. The problem becomes more difficult when several vehicles approach within similar time intervals.

A recent review by Al-Sharman et al. identifies unsignalized-intersection navigation as a particularly challenging autonomous-driving problem because it combines uncertainty, interaction among multiple agents, partial observations, and safety-critical decision-making. The review concludes that although many learning and optimization methods have been proposed, a robust and practical solution for real-world unsignalized-intersection navigation remains an open problem [1].

An important distinction is therefore required between **proximity** and **conflict**. Vehicles may be physically close but have compatible paths, while vehicles arriving from different approaches may share a crossing or merging region. Effective intersection coordination therefore requires reasoning about intended paths and conflict geometry rather than considering only inter-vehicle distance.

This distinction directly motivates the use of explicit movement paths and conflict-zone reasoning in the present research.

2.2.2 Autonomous Intersection Management

Autonomous Intersection Management (AIM) refers to approaches in which connected or autonomous vehicles coordinate their movement through intersections without relying solely on conventional traffic signals.

One of the earliest influential approaches was proposed by Dresner and Stone. Their reservation-based AIM model treats vehicles as autonomous agents that request space-time reservations from an intersection manager. The manager accepts or rejects requests based on whether the proposed trajectories conflict with previously accepted reservations [9]. This work demonstrated that vehicle-level communication could potentially support much finer control than conventional signals. However, the approach depends on a dedicated intersection manager, making the decision process fundamentally centralized.

Centralized coordination has continued in more recent learning-based work. Guillen-Perez and Cano proposed a multi-agent deep reinforcement-learning architecture for Connected Autonomous Vehicles in which intersection-level coordination is used to control vehicle speeds. Their results demonstrated substantial improvements in travel-time-related measures compared with conventional approaches [4]. However, a common decision structure still coordinates the participating vehicles rather than allowing each vehicle to independently establish its runtime permission.

Aoki and Rajkumar proposed the Cyber Traffic Light, in which vehicles cooperate at dynamic conflict regions through V2V communication. The work is important because it recognizes that conflict-management mechanisms may need to appear dynamically rather than exist only at permanent signalized intersections. Nevertheless, the protocol effectively constructs a traffic-management role responsible for coordinating access to the conflict region [10].

These approaches establish an important result: communication between intelligent vehicles can improve intersection management. At the same time, they highlight the distinction between communication-enabled coordination and fully decentralized execution. V2V communication alone does not make a system decentralized if a central or temporary manager still determines the final ordering for all vehicles.

2.2.3 Partial Observability and Intention Uncertainty

An autonomous vehicle does not necessarily know the future maneuver of another road user when the interaction begins. At an intersection, the difference between a vehicle travelling straight and turning left or right can completely change the associated conflict relationships.

For this reason, intention prediction is closely related to intersection decision-making. Vehicle history, speed, acceleration, heading, lateral motion, and contextual features can provide information about future maneuvers before the maneuver itself becomes fully observable.

The inD dataset was introduced by Bock et al. to provide naturalistic trajectory data from urban intersections. It contains more than 13,500 road users and approximately ten hours of data recorded at four German intersections. The drone-based collection method provides trajectory information with substantially reduced ground-level occlusion and has been used for prediction, behavior modelling, and scenario-based validation research [8].

Sequence-learning methods are particularly relevant because intention is expressed through changes in motion over time rather than through a single observation. Previous research has demonstrated the use of recurrent neural networks for predicting intersection maneuvers. For example, Phillips et al. showed that LSTM-based models could predict whether a driver would turn left, turn right, or travel straight before reaching an intersection [18]. Other work has combined maneuver recognition with LSTM-based future-trajectory prediction for conflict detection at unsignalized intersections [17].

The implication for right-of-way negotiation is important. If the movement intention of another vehicle is uncertain, then the exact conflict relationship may also be uncertain. A coordination architecture therefore needs to distinguish between confirmed physical relationships and relationships that depend on uncertain intention estimates. Treating unknown intentions as known ground truth could make the decision process unrealistically optimistic.

2.2.4 Multi-Agent Reinforcement Learning and CTDE

Reinforcement learning provides a mechanism through which an agent can improve a policy through repeated interaction with the environment. In a multi-agent setting, however, each agent's environment changes partly because other agents are also learning and acting.

This creates the problem of **non-stationarity**. From the perspective of a single learner, the behavior of other agents changes during training, which makes learning more difficult than in a single-agent Markov Decision Process. Lowe et al. addressed this issue using centralized critics that have access to additional information during training while retaining distributed actors [14]. This idea became an important foundation for **Centralized Training with Decentralized Execution (CTDE)**.

MAPPO applies PPO-style policy optimization to cooperative multi-agent settings. Yu et al. demonstrated that PPO-based multi-agent methods can perform competitively across several cooperative benchmarks when implemented carefully [3]. MAPPO is therefore relevant to intersection negotiation because agents can be trained cooperatively while still executing their policies separately after training.

However, CTDE does not by itself guarantee that runtime execution is decentralized in every part of a system. A system can use decentralized learned actors but later pass their outputs to a central planner that determines the final physical commands. Consequently, the architecture after the learned decision layer is also important when assessing whether a complete system is decentralized.

This issue influenced the final design of the present research, where the learned actors operate without the centralized critic at runtime and each ego vehicle subsequently computes its own physical permission from the coordination information relevant to that vehicle.

2.2.5 Constrained Learning and Invalid Actions

A reinforcement-learning policy operating in a safety-critical domain should not be allowed to select every mathematically representable action if some actions are semantically invalid or violate non-negotiable constraints.

Invalid-action masking is one technique for enforcing state-dependent action availability in policy-gradient methods. S.Huang and S.Ontañón showed that masking invalid actions can be represented as a valid policy-gradient operation and can become particularly important when the action space contains many choices that are impossible in a particular state [15].

The same principle is relevant to right-of-way negotiation. For example, a vehicle should not be permitted to “negotiate away” an obligation if the traffic-rule layer identifies the obligation as mandatory and non-negotiable. Similarly, a responder should not respond to a proposal that does not exist or that belongs to another claim.

Therefore, an important theoretical distinction in this research is between:

- the policy's available semantic choices, and
- the complete physical set of possible vehicle actions.

The policy in the final framework does not select steering or acceleration directly. It selects from a constrained negotiation vocabulary, while hard masks and deterministic reasoning determine which actions are legally and semantically available.

2.2.6 Safety, Liveness, Efficiency and Fairness

Collision avoidance is necessary but is not sufficient for successful intersection coordination. A system could avoid every collision by holding every vehicle indefinitely. Such behavior would be safe from one perspective but useless as an intersection-management strategy.

Accordingly, autonomous-intersection research commonly evaluates both safety and traffic performance. Measures include collisions, average travel time, delay, throughput, waiting time, and completion.

Fairness is also important, particularly under congestion. Wu et al. showed that optimizing only for efficiency at an autonomous intersection can negatively affect fairness. Their work measures fairness through the difference between vehicle approach order and crossing order, while traffic efficiency is represented through average travel time. FCFS is included as a fairness-oriented benchmark [13].

This is relevant to the present project because average traffic performance alone can hide starvation. An approach may experience a reasonable overall mean delay while one individual vehicle or one direction waits substantially longer than the others.

Therefore, the final evaluation distinguishes:

Safety - whether unsafe physical events occur.
Liveness - whether vehicles make progress and complete their routes.
Efficiency - how much time is required and how much traffic is discharged.
Fairness - whether delays or service order are unevenly distributed.

This distinction is particularly important when analyzing conservative safe-hold behavior.

2.3 Related research

2.3.1 Reservation and Centralized Coordination Approaches

The reservation-based AIM framework developed by Dresner and Stone is a foundational example of multi-agent intersection coordination. Vehicles communicate their desired crossing trajectories to a central intersection manager, which reserves conflict-free regions of space and time [9]. The framework demonstrated that individual vehicle scheduling can potentially use intersection capacity more efficiently than conventional signals. Its principal limitation relative to the present research is architectural: the final decision is produced by a central intersection agent.

Later approaches extended centralized intersection control using reinforcement learning. The RAIM architecture proposed by Guillen-Perez and Cano applies multi-agent deep reinforcement learning to coordinate CAV speeds at intersections [4]. The learning-based mechanism provides adaptability beyond fixed control rules, but the system still depends on an intersection-level control architecture.

Feng et al. proposed a right-of-way-based multi-agent deep reinforcement-learning framework for dynamic unsignalized intersections [6]. Their work is particularly relevant because it explicitly integrates real right-of-way relationships into the learning process. A Right-of-Way Cooperator limits the learning decision space using vehicle priority and conflict detection, while the learning algorithm handles collaborative decisions. The work also demonstrates that pure reinforcement learning cannot simply be assumed to learn full traffic-rule compliance.

This provides strong motivation for the hybrid design used in the present thesis: deterministic precedence is treated as a constraint on learning rather than as behavior the agent is expected to rediscover entirely through reward.

2.3.2 Decentralized Multi-Agent Coordination

Wu et al. proposed DCL-AIM, a decentralized coordination-learning method for autonomous intersection management [5]. Rather than treating every vehicle interaction as one monolithic high-dimensional decision, DCL-AIM decomposes the state into independent and coordinated components and dynamically identifies where

coordination is needed. Its evaluation reported improved performance compared with FCFS, longest-queue-first, and signal-control benchmarks.

This idea is highly relevant to the present research because not every vehicle near an intersection needs to participate in the same coordination decision. Vehicles with non-conflicting paths should not unnecessarily constrain one another, while vehicles that share a relevant conflict region require stronger coordination.

Zhuang et al. proposed a decentralized attention-based MAPPO method for mixed traffic at an unsignalized intersection [11]. Their model uses an attention module to extract relevant neighboring information and combines traffic rules, trajectory prediction, and cooperative MARL. The results indicated improved adaptability compared with a heuristic rule-based method.

This work demonstrates two important ideas that are also relevant to the present thesis: decentralized multi-agent decision-making can be practical, and not all neighboring vehicles should contribute equally to an agent's decision.

However, the present research uses different decompositions. Instead of using learned attention to decide which neighbors are important, it first uses geometric and temporal conflict analysis to restrict coordination to physically relevant relationships.

2.3.3 Graph-Based Interaction Modelling

Intersection interactions naturally form relational structures. Vehicles can be represented as nodes, while conflict, precedence, communication, or yielding relationships can be represented as edges.

Zhou et al. proposed a Reasoning Graph-based Reinforcement Learning framework for mixed traffic at unsignalized intersections [12]. Their model represents interactions using graphs and employs Graph Convolutional and Graph Attention networks to capture dependencies and chain reactions among vehicles. Their evaluation showed improvements in conflict-related and travel-time measures, and the work further included hardware-in-the-loop and vehicle-in-the-loop validation.

The use of graphs is particularly suitable for right-of-way reasoning because precedence is naturally directional. A statement such as vehicle A must yield to

vehicle B cannot be represented adequately by an undirected proximity relation. It is a direct relationship.

The current research therefore uses directed precedence graphs to represent right-of-way claims. However, unlike approaches where the graph directly represents a globally determined crossing sequence, each vehicle first constructs its own local view and coordination graph from ego-local observations and structured messages.

2.3.4 Game-Theoretic and Behavior-Aware Coordination

Game theory provides another framework for modelling strategic interaction among road users.

Chandra and Manocha proposed GamePlan, an auction-based game-theoretic method for planning among human drivers and autonomous vehicles at intersections, roundabouts, and merging areas [7]. The approach considers driving style and assigns different priorities to aggressive and conservative drivers. The authors show that considering behavioral differences can improve coordination compared with simpler time-based auctions.

The strength of this type of approach is that interaction is treated explicitly rather than assuming every road user follows an identical deterministic policy. However, the negotiation model is different from the local claim-based learned negotiation used in this thesis.

A related concept is learned courteous behavior. Yan et al. trained an RL controller that allowed automated vehicles to voluntarily relinquish right-of-way to improve intersection traffic flow [16]. This work provides evidence that relinquishing a priority claim can be a meaningful learned coordination behavior rather than simply a failure to assert priority. However, their approach uses a centralized controller, and voluntary yielding in a learning model should not be interpreted automatically as legally equivalent to a particular human right-of-way communication rule.

This distinction is important in the present framework. Relinquishment is treated as a computational negotiation action that requires an explicit corresponding response; it is not itself a physical braking command or unconditional permission for another vehicle to enter the intersection.

2.3.5 Intention-Aware Conflict Reasoning

Many intersection decision-making methods assume that the intended trajectory of surrounding vehicles is already known or communicated. In realistic partial-observation settings, this assumption may not hold.

The inD dataset provides naturalistic urban-intersection trajectories that can support prediction of vehicle movement before the maneuver is fully completed [8]. Sequence models such as LSTM and GRU are suitable for this type of problem because turning intent is expressed through temporal patterns rather than a single frame.

Phillips et al. demonstrated generalizable intention prediction at intersections using LSTM models, reporting prediction of left, right, and straight movements across different intersection types [18]. Cao et al. similarly demonstrated that recognizing turning behavior and using recurrent networks for trajectory prediction can support conflict detection at unsignalized intersections [17].

These works support the inclusion of an intention-prediction stage before right-of-way reasoning. In the implemented project, real-world inD trajectories are used to train the intention model, while system-level negotiation remains evaluated in SUMO. This preserves the distinction between real-world data support for perception-related inference and simulation-based evidence for the right-of-way system.

2.4 Existing systems products and techniques

Existing autonomous intersection techniques differ primarily in where decisions are made, what information is required, and how safety is enforced.

Rule-based approaches provide deterministic behavior and high interpretability but can become conservative when multiple valid or ambiguous interactions occur. FCFS is simple and usually easy to explain, but it may unnecessarily serialize vehicles whose paths do not conflict.

Reservation-based AIM improves intersection utilization by scheduling specific space-time regions. However, reservation systems generally depend on an intersection manager and require accurate future trajectory information.

Centralized learning approaches can optimize decisions across several vehicles using global or infrastructure-level information. Their advantage is access to a broad view of the traffic state, but they create a different deployment assumption from vehicle-owned decentralized execution.

Decentralized MARL approaches remove some dependence on central decision authorities. However, they still need mechanisms for handling partial observability, non-stationarity, safety constraints, and inconsistent local beliefs.

Game-theoretic and auction-based techniques explicitly model strategic interaction, but their coordination process is often based on a particular bidding or behavioral model.

Hybrid rule-and-learning approaches are increasingly important because they combine the adaptability of learning with externally defined safety or regulatory constraints. Feng et al.'s right-of-way cooperator and Zhuang et al.'s integration of traffic rules with MAPPO are examples of this direction [6], [11].

Table 2 : Comparison of Representative Intersection Coordination Approaches

Study	Approach	Control	Key Strength	Main Limitation
Dresner & Stone [9]	Reservation-based AIM	Centralized	Efficient space-time scheduling	Requires an intersection manager
Guillen-Perez & Cano [4]	Multi-agent DRL	Centralized	Improve traffic efficiency	Depends on central coordination
Wu et al. [5]	Decentralized coordination learning	Decentralized	Coordinates only relevant agents	Limited explicit rule/intention integration
Chandra & Manocha [7]	Game-theoretic bidding	Distributed	Models interactive driving behavior	Uses auction-style one-time priority
Zhuang et al. [11]	Attention-based MAPPO	Decentralized	Learn cooperation from nearby traffic	Different interaction/safety structure
Feng et al. [6]	Right-of-way-based MARL	Centralized	Combines learning with traffic rules	Final coordination remains centralized
This research	Rules + conflict reasoning + MAPPO	Decentralized	Combines local reasoning, rules and learned negotiation	Limited to selected four-way SUMO scenarios

2.4.1 Simulation and Evaluation Techniques

SUMO is widely used in intersection-management research because it allows reproducible microscopic traffic simulation together with programmatic vehicle control. Several of the studies reviewed above evaluate their methods using simulation rather than direct public-road deployment [5], [11], [16].

This is appropriate for early-stage autonomous-intersection research because scenarios can be repeated under controlled conditions and unsafe behaviors can be studied without physical risk. However, simulation success cannot be interpreted as proof of real-world deployment safety.

The reviewed literature also shows that evaluation should not depend on a single successful demonstration. Common metrics include collision behavior, travel time, delay, average speed, throughput, waiting time, and deadlock-related behavior. More recent work additionally highlights fairness between vehicles [13].

These observations informed the evaluation framework adopted in this thesis, which reports both successful and degraded cases instead of considering collision avoidance alone as sufficient evidence of system success.

2.5 Gap synthesis and conceptual framework

The reviewed literature shows that no single missing algorithm defines the research gap. Multi-agent reinforcement learning has already been used at unsignalized intersections. Decentralized coordination has also been studied. Graph-based interaction models, right-of-way constraints, intention prediction, and fairness metrics each have substantial prior research.

The more specific gap identified through this review concerns the integration and separation of these functions.

Several centralized AIM systems achieve effective traffic control but depend on an intersection-level decision authority [4], [6], [9]. Decentralized learning methods reduce this dependency but do not necessarily combine explicit regulatory precedence with uncertain movement intentions and vehicle-owned physical execution [5], [11]. Game-theoretic systems model strategic interaction but operate using different bidding and behavioral assumptions [7]. Intention-prediction research estimates future movement, but prediction alone does not establish right-of-way [8], [17], [18]. Graph-based methods represent complex interactions effectively but often use the graph as part of a centralized or sequential cooperative decision process [12].

The final research problem is therefore formulated around the following combination:

1. Each autonomous vehicle begins with an ego-local and partially observable view of the intersection.
2. Movement intention is estimated rather than assumed to be perfectly known.
3. Map geometry determines which candidate vehicle paths conflict.
4. Temporal reasoning removes conflicts that are physically irrelevant at the current time.
5. Mandatory traffic-rule precedence is evaluated before reinforcement learning.
6. Only explicitly negotiation-eligible relationships are passed to the MAPPO policy.

7. The learned action modifies semantic right-of-way claims rather than directly selecting vehicle acceleration or steering.
8. A deterministic protocol verifies agreement and maintains an auditable effective precedence representation.
9. Each vehicle independently converts the coordination result into its own physical conflict-zone permission.
10. Evaluation measures safety, liveness, efficiency, fairness, and failure behavior separately.

This separation is central to the contribution of the thesis. Learning is used where adaptation is needed, but mandatory regulatory and physical relationships are not delegated entirely to the learned policy.

Table 3 : Literature gap and research response

Gap identified from literature	**Response in this research**
Central intersection managers remain common	Runtime physical permission is computed by each ego vehicle
Pure learning may not guarantee rule compliance	Mandatory precedence is established outside MAPPO
Full trajectory/intention often assumed known	Learned intention estimation supports local conflict reasoning
Nearby vehicles are sometimes treated uniformly	Map and temporal conflict reasoning identify relevant relationships
RL may produce semantically invalid actions	State-dependent hard action masks constrain negotiation
Negotiation output may be difficult to interpret	Actions are explicit claim semantics: keep, relinquish, accept, reject
Collision-free behavior may still deadlock	Safety and liveness are evaluated separately
Efficiency metrics can hide unfair service	Per-vehicle and per-approach waiting/fairness metrics are included
Dense traffic creates scalability challenges	Continuous four-approach traffic and failure cases are retained in evaluation

The resulting conceptual framework can be understood as a staged reduction of uncertainty and freedom to decide. Local observations first establish what is believed about the surrounding traffic. Intention prediction narrows possible maneuvers. Conflict and temporal reasoning identify which interactions are physically relevant. Traffic rules then determine which relationships are mandatory. Only the remaining eligible relationships are exposed to learned negotiation. Finally, each vehicle independently interprets the valid coordination commitments for physical execution.

This framework differs from an end-to-end learned controller. The research does not attempt to learn every stage from raw observations to acceleration. Instead, it investigates whether a constrained learned negotiation mechanism can operate within a larger, traceable decentralized right-of-way architecture.

Literature therefore supports the central premise of this thesis: neither rigid rules nor unconstrained reinforcement learning alone is sufficient for the selected problem. A hybrid design offers a way to preserve explicit constraints while still allowing adaptive negotiation where the deterministic layers do not provide a complete coordination result.

3 RESEARCH METHODOLOGY

3.1 Research design

This research followed design-and-experimental methodology. The work involved developing a decentralized right-of-way negotiation framework and evaluating its behavior in a controlled simulation environment. The study was not designed as a real-road deployment trial; instead, the objective was to investigate whether the proposed combination of local perception, intention estimation, traffic-rule reasoning, conflict analysis, and learned negotiation could operate coherently within a reproducible four-way intersection scenario.

The research was carried out in three main stages. First, the perception and reasoning pipeline was established. Each autonomous vehicle maintained an ego-local representation of nearby traffic, estimated the likely maneuvers of relevant vehicles, and identified spatial and temporal conflicts. Mandatory right-of-way relationships were then derived using the selected traffic-rule profile.

Second, a learned negotiation mechanism was developed for cases that remained eligible for negotiation. Multi-Agent Proximal Policy Optimization (MAPPO) was selected because it supports centralized training while allowing actors to operate using local observations during execution [3]. Invalid or protocol-inconsistent negotiation actions were removed through state-dependent action masks, following the general principle of invalid-action masking in policy-gradient methods [15].

Finally, the completed framework was evaluated in SUMO under rule-resolved, negotiation-required, complex, congested, and failure-oriented traffic cases. Safety, liveness, efficiency, fairness, and runtime-integrity evidence were recorded separately. This distinction was important because a collision-free run could still represent an unsuccessful outcome if vehicles remained indefinitely blocked.



Figure 1 : Research methodology workflow

3.2 Data participants or study materials

3.2.1 Simulation Environment

The system-level experiments were conducted using Simulation of Urban MObility (SUMO) with TraCI used to exchange simulation state and control commands with the Python implementation. SUMO provides microscopic, time-discrete traffic simulation together with a remote-control interface, making it suitable for controlled vehicle-interaction experiments [19].

The selected environment was an unsignalized four-way intersection with one incoming traffic stream from each of the north, east, south, and west approaches. Each approach supported left-turn, straight, and right-turn movements, giving 12 legal movement paths. Geometric analysis of these paths produced 30 physical conflict zones used by the conflict-reasoning layer.

The simulation step was fixed at **0.04 s**. Autonomous vehicles used a maximum speed of **13.89 m/s**, length of **5.0 m**, width of **1.8 m**, acceleration of **2.0 m/s²**, and normal deceleration of **4.5 m/s²**. SUMO's native vehicle-safety behavior remained active during physical execution.

Table 4 : Main experimental configuration

Parameter	**Value**
Simulator	SUMO + TraCI
Intersection	Unsignalized four-way
Simulation step	0.04 s
Legal movements	12
Physical conflict zones	30
Maximum vehicle speed	13.89 m/s
Vehicle length / width	5.0 m / 1.8 m
Acceleration / deceleration	2.0 / 4.5 m/s²
Traffic type	Autonomous vehicles only
Maximum simultaneous MAPPO participants	4

3.2.2 Perception and Local Observation

Runtime decision-making was based on an ego-local observation abstraction rather than direct access to the complete SUMO world state. Each autonomous vehicle maintained a Local Dynamic Map (LDM) containing locally observed vehicles and their recent motion histories.

The default experimental perception profile used geometric range, field-of-view, visibility, and occlusion reasoning. A four-sensor reference configuration was modelled with a **160 m individual sensing range** and fused surround coverage. This was an object-level simulation abstraction and was not treated as a validated physical radar model.

A conservative observation-distance check was also included in the implementation. The project used a 2.0 s motion-history window, a 1.0 s primary prediction horizon, a maximum approach speed of 13.89 m/s, and a project-specific 35 m safety margin. These values were used to verify that the selected observation range was sufficient for the defined experimental context. The 35 m margin is an engineering assumption of this study rather than a value prescribed by a traffic standard or sensor manufacturer.

3.2.3 Intention-Prediction Data

Real-world trajectory data was used specifically for the movement-intention component. The **inD dataset** contains naturalistic trajectories captured at German intersections and was selected because it provides high-resolution intersection behaviour suitable for manoeuvre prediction [8].

Recordings 7-14 were used for training, recordings 15-16 for validation, and recording 17 for testing. Each sample was derived from a nominal 2.0 s history sampled at 25 Hz. From the 50 recorded positions, causal backward differences produced a 48 × 6 input sequence containing speed, acceleration magnitude, longitudinal and lateral velocity, and longitudinal and lateral acceleration.

Two GRU models were deployed:

- a primary model predicting intention 1.0 s ahead; and

- a secondary model predicting intention 0.5 s ahead.

The three output classes were LEFT, RIGHT, and STRAIGHT. The final models were exported to ONNX and executed using CPU-based ONNX Runtime in the SUMO system.

Table 5 : Intention-model data and input design

Item	**Configuration**
Dataset	inD
Training recordings	7–14
Validation recordings	15–16
Test recording	17
Sampling frequency	25 Hz
Observation history	2.0 s
Model input	48 × 6 causal features
Primary prediction horizon	1.0 s
Secondary prediction horizon	0.5 s
Classes	Left / Right / Straight
Deployment format	ONNX, CPU inference

3.3 Experimental or study procedure

The experimental procedure followed the same processing order used by the final runtime system.

At each simulation step, vehicle states were obtained from SUMO and transformed into ego-local observations. Each vehicle updated its own LDM and generated an intention estimate for relevant surrounding vehicles. The predicted movement was then associated with one of the predefined intersection paths.

Potential interactions were filtered using path geometry. Vehicles whose candidate paths did not intersected were excluded from further right-of-way reasoning. For vehicles with shared conflict regions, temporal occupancy reasoning estimated whether the conflict was currently relevant.

Mandatory precedence was evaluated before learned negotiation. The experimental intersection used the German uncontrolled equal-priority road context represented by the project profile DE_STVO_UNCONTROLLED_4WAY_V1. The German StVO §8 specifies that, at equal-priority intersections, traffic approaching from the right has priority unless another priority rule applies. Turning-related precedence was additionally handled by the project's regulatory layer.

Where rules provided a clear non-negotiable precedence relationship, MAPPO was not invoked. Only relationships identified as negotiation-eligible were represented in the learned negotiation graph.

3.3.1 Negotiation Training and Model Selection

Negotiation scenarios were generated from the legal movement catalogue and traffic-rule precedence graphs. The catalogue contained **108 regulatory-cycle scenario specifications**, which were divided deterministically into **36 training, 36 validation, and 36 held-out scenarios** with not pairwise overlap.

MAPPO training followed the Centralized Training with Decentralized Execution principle. While learning, the critic could use the permitted joint training representation, while the deployed actors used local/joint-local information only. MAPPO is well suited to this setting because centralized value estimation can assist

cooperative learning without requiring the critic during final decentralized execution [3].

The graph representation used 8-dimensional node features, 9-dimensional edge features, a hidden dimension of 64, and three message-passing layers. Separate role-specific heads were used for proposer and responder actions. The semantic actions were:

Proposer: KEEP_CLAIM or RELINQUISH_CLAIM
Responder: ACCEPT_RELINQUISHMENT or REJECT_RELINQUISHMENT

Hard Boolean masks removed actions that were impossible or invalid for the current protocol state. This follows the established use of invalid-action masking to restrict policy-gradient agents to feasible state-dependent actions [15].

A bounded one-factor model-selection experiment compared 5, 10, and 15 PPO update epochs (E5, E10, and E15). Other configuration values were held constant, including a learning rate of 0.0005, PPO clipping parameter of 0.2, three message-passing layers, and the frozen GNN representation. Three canonical replications were used for each candidate. Candidate selection was based on the lowest mean validation Total Team Travel Time (TTT) among configurations satisfying the predefined safety and integrity gates. Held-out scenarios were not used for parameter selection.

The selected policy was frozen before subsequent system validation; no retraining was performed during panel evaluation.

3.3.2 System-Level Validation

The completed system was evaluated using targeted cases chosen to cover different types of behavior rather than only a single successful demonstration.

The compact validation included:

- a rule-resolved two-vehicle case;
- a regulatory-cycle case;
- a multiple-left-turn complex case;

- continuous four-approach traffic with 16 scheduled vehicles;
- a simultaneous four-way critical case; and
- controlled missing/stale coordination-message checks.

The congested case used four vehicles per approach and represented all twelve legal movements. Although total live traffic could exceed four vehicles, only the physically relevant queue-headed vehicle from each approach could participate in a negotiation event. This preserved the trained maximum of four MAPPO participants while allowing followers to remain in normal SUMO car-following behavior.

The evaluation retained unsuccessful outcomes rather than adjusting the scenarios until they passed. Runs were classified as **PASS**, **SAFE_DEGRADATION**, **UNSAFE_FAILURE**, or **INVALID_EXPERIMENT**, depending on completion, safety, and experimental-integrity conditions.

3.4 Measures and evaluation metrics

The evaluation deliberately separated **safety** from **liveness**. A stopped vehicle may avoid collision but still create deadlock; therefore, collision count alone was not used as evidence of successful intersection management.

Table 6 : Evaluation measures

Category	Main measures
Safety	Collisions, blocked-zone violations, emergency-braking evidence
Liveness	Completion rate, unfinished vehicles, deadlock/timeout
Efficiency	Total team travel time, mean travel time, throughput
Waiting	Mean, P95 and maximum waiting time
Fairness	Per-approach means wait, approach waiting-time gap, FCFS inversion rate
Negotiation	MAPPO calls, accepted/rejected outcomes, fallback activations
Scalability	Active/moving vehicles, queue depth, max negotiation participants
Integrity	Policy unchanged, critic calls = 0 at runtime, no training during evaluation

Waiting time was accumulated when a vehicle's speed was below **0.5 m/s** while it remained on an incoming lane. Throughput was calculated as the number of vehicles that completed the network per observed simulated minute.

Fairness was assessed using the mean waiting time for the north, east, south, and west approaches and the difference between the highest and lowest approach mean. The service-order inversion measure was included because previous work has shown that intersection efficiency alone can hide unfair treatment between vehicles [13].

No single weighted “overall fairness score” was introduced.

3.5 Data analysis

The intention-prediction models were evaluated using accuracy, balanced accuracy, macro precision, macro recall, macro F1-score, and log loss. Class-level results for left, right, and straight movements were also retained. The final system evaluation used the measures described in Table 6.

For repeated model-selection experiments, mean validation Total Team Travel Time was used to compare the predefined E5, E10, and E15 configurations. This comparison was used only to select the best eligible configuration among the tested candidates and was not interpreted as proof of global optimality.

For system-level validation, each scenario was analyzed according to both its safety outcome and its ability to complete traffic demand. Runs with no collision or blocked-zone violation but incomplete traffic discharge was retained as **safe degradation** rather than being reported as successful runs.

The study therefore used descriptive analysis rather than making broad statistical claims that were not supported by the bounded experimental design.

3.6 Ethics privacy security and responsible AI

The study did not involve direct interaction with human participants. The inD dataset contains previously collected, anonymized trajectory information made available for research purposes [8]. No, personally identifying information was collected as part of this research.

All final right-of-way experiments were performed in simulation, avoiding physical risk to road users. Nevertheless, the system was treated as safety-critical research. Learned decisions were therefore constrained by deterministic traffic-rule reasoning, conflict-zone checks, hard action masks, and SUMO's native physical-safety mechanisms.

The simulated V2V communication mechanism represents structured coordination information rather than a production communication network. Wireless attacks, packet-loss distributions, authentication, and cryptographic security were outside the scope of the study. Missing and stale coordination information were instead tested as bounded fail-safe conditions.

Generative AI and automated coding tools used during development and thesis preparation must be disclosed separately in **Appendix B**, as required by the current individual-thesis template.

3.7 Reproducibility and research artefacts

Reproducibility was supported by keeping simulation configurations, movement paths, scenario manifests, model checkpoints, experimental results, and validation reports within the project repository.

Training, validation, and held-out scenario groups were frozen before final MAPPO selection. The final selected checkpoint was identified by a SHA-256 hash, and validation checks confirmed that the checkpoint remained unchanged during evaluation. Runtime integrity monitoring also verified that no optimizer, backward pass, or centralized critic call was performed during deployed evaluation.

The final project retains:

- SUMO network and route definitions;
- intention-model ONNX files and feature contracts;
- MAPPO scenario manifests and selected checkpoints;
- automated unit and integration tests;
- panel-validation scenario manifests;
- per-run evidence and replay information; and
- generated research reports.

This separation between training artefacts and post-training evaluation evidence reduces the possibility of unintentionally modifying the policy while producing results.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The final design requirements were derived from the research problem, the defined four-way intersection scope, and the need to separate mandatory safety constraints from learned negotiation. The component was not intended to act as an end-to-end driving controller. Instead, it was designed to determine and execute right-of-way decisions within a controlled autonomous-vehicle intersection environment.

The main requirements are summarized in Table 7.

Table 7 : Main design requirements

	Requirement	Design response
R1	Decisions should use ego-local runtime information	A separate LDM and reasoning context is maintained for each AV
R2	Vehicle maneuvering uncertainty should be considered	GRU-based intention prediction estimates left, right, or straight movement
R3	Only physically relevant vehicles should influence coordination	Movement paths, conflict zones, and temporal occupancy are checked before negotiation
R4	Mandatory right-of-way rules must not be overridden by learning	Traffic-rule precedence is evaluated before MAPPO
R5	Learned actions must remain interpretable and constrained	MAPPO uses semantic keep/relinquish and accept/reject actions with hard masks
R6	Runtime execution should remain decentralized	Each ego vehicle independently computes its own permission and speed constraint
R7	Unsafe or incomplete coordination should fail conservatively	Missing, stale, or physically unordered coordination leads to a safe hold
R8	Congested traffic should not require a larger learned agent graph	Only physical queue heads negotiate; followers remain under SUMO car-following
R9	Experiments should be auditable and reproducible	Policies, manifests, scenario definitions, and evaluation evidence are frozen and versioned

The requirements also reflect an important distinction between **safety** and **liveness**. A vehicle must not enter an unresolved conflict simply to improve traffic flow, but a design that always stops vehicles is also not an effective intersection-management solution. The system therefore attempts to preserve safety first while allowing liveness and efficiency to be measured separately.

4.2 Component architecture within the complete solution

Multi-Agent Negotiation for Right-of-Way in Complex Intersections is one of four independently developed research components within the wider Agentic AI for Robust and Socially-Aware Autonomous Driving project. The components share the same overall research theme, but the current implementation does not form a single live algorithmic pipeline in which the output of one component is automatically consumed by the next. The unified showcase provides a common presentation layer, while each component retains its own implementation and experimental evidence.

Accordingly, Component 2 contains its own intention-estimation pipeline for vehicle maneuvers and does not depend on the pedestrian-intention model developed in Component 1. This separation is maintained in the thesis so that the individual research contribution and its evidence remain clear.

Within Component 2, the processing pipeline begins with SUMO vehicle states. Each vehicle creates an ego-local observation and updates its Local Dynamic Map. The deployed GRU models estimate relevant vehicle maneuvers, after which movement paths and conflict zones are used to identify physically and temporally relevant interactions. Mandatory precedence is then derived from the traffic-rule layer.

Each vehicle publishes structured precedence claims rather than its complete private observation. These claims are used to reconstruct an ego-owned joint-local negotiation graph. The graph is encoded using an edge-aware message-passing representation, after which the MAPPO proposer and responder actors select only among actions allowed by the current hard action masks. Message-passing neural networks provide a general mechanism for learning or constructing representations from relational graph structures [20].

A deterministic negotiation protocol then interprets the selected semantic actions. The resulting commitments are not treated as direct vehicle-control commands. Each ego vehicle independently checks the physical relevance of its commitments and determines whether it should proceed or remain held before the relevant conflict zone.



Figure 2 : Final architecture of the decentralized right-of-way negotiation component

The choice of centralized training with decentralized execution follows the general CTDE principle used in cooperative multi-agent learning [3], [14]. However, the design extends decentralization beyond the learned actor: the final physical permission is also computed locally rather than being reconstructed by a centralized runtime planner.

4.3 Design alternatives and decisions

Several alternatives were considered during the development of the final architecture. The most important decisions are summarized in Table 8.

Table 8 : Key design alternatives and final decisions

Design issue	Alternative	Final decision	Main reason
Runtime coordination	Central intersection manager	Ego-local decentralized execution	Avoid dependence on a single runtime decision authority
Decision strategy	Pure rules or pure RL	Hybrid rules + MAPPO	Preserve mandatory precedence while allowing adaptive negotiation
Learned output	Direct acceleration/speed	Semantic right-of-way action	More interpretable and easier to constrain
Interaction representation	All nearby vehicles	Conflict-relevant joint-local graph	Avoid unnecessary agent interactions
RL action space	Unrestricted actions	Hard state-dependent masks	Prevent invalid or non-negotiable actions
Dense traffic handling	One actor for every live vehicle	Queue-head negotiation	Keep learned context within the trained four-agent scope
Graph representation	Flat fixed-order vector	Edge-aware graph representation	Preserve directed vehicle relationships

A centralized intersection manager can coordinate traffic efficiently because it can observe several vehicles jointly. Reservation-based and centralized learning systems have demonstrated the advantages of this approach [4], [9]. The aim of this research,

however, was to investigate vehicle-owned decision-making. Decentralized coordination research such as DCL-AIM and attention-based MAPPO provides evidence that coordination can be organized without a permanent centralized runtime manager [5], [11].

The project also moved away from the proposal-stage idea of allowing reinforcement learning to determine most right-of-way behavior directly. Previous work has shown that learned intersection policies may still require explicit right-of-way constraints to improve regulatory compliance [6]. For this reason, the final design treats mandatory precedence as a hard input to negotiation rather than as behavior that MAPPO must learn through reward alone.

A second important decision was to make learned actions semantic rather than physical. MAPPO does not select steering angles, acceleration, braking force, or target speed. Instead, an eligible proposer may keep or propose relinquishing a precedence claim, while the responder may accept or reject that proposal. This provides a clearer boundary between learned negotiation and physical safety control.

Hard action masking was also adopted so that the policy cannot choose an action that is invalid in the current protocol state. State-dependent invalid-action masking is consistent with policy-gradient methods and is useful where only part of an action vocabulary is feasible in a given state [15].

4.4 Detailed design

4.4.1 Ego-Local Perception and Local Dynamic Map

The system maintains a separate runtime view for each autonomous vehicle. SUMO provides the underlying simulated vehicle states, but the actor does not receive the complete simulator state as a perfect global observation.

For each ego vehicle, the perception layer filters the observable traffic according to the configured sensing abstraction. The resulting objects are maintained in an ego-specific Local Dynamic Map (LDM) together with recent motion history. The LDM therefore provides a persistent local representation rather than treating each SUMO step as an unrelated snapshot.

This design allows different vehicles to hold different local views of the same intersection, which is important for investigating decentralized coordination under partial observation.

4.4.2 Movement-Intention Estimation

The intention layer estimates whether relevant observed vehicles are likely to move **left, right, or straight**. The deployed models use GRU sequence models exported to ONNX and operate on recent motion histories.

Intention prediction is treated as supporting evidence rather than perfect ground truth. This is important because the physical conflict between two vehicles depends strongly on their intended movements. Previous intersection-intention studies have similarly shown that temporal motion histories can support early maneuver prediction [17], [18].

The intention output is then associated with one of the predefined movement paths of the intersection. This creates the link between learned prediction and the deterministic geometry used by the next stage.

4.4.3 Movement Paths and Conflict-Zone Reasoning

The intersection contains four incoming approaches, each supporting a left turn, straight movement, and right turn. This produces **12 legal movement paths**.

The paths are generated from the SUMO network and compared geometrically to identify areas where movements cross or merge. The final map contains **30 physical conflict zones**. Two vehicles therefore do not become negotiation partners simply because they are nearby; their candidate movement paths must first create a relevant physical relationship.

Temporal occupancy is then considered. If two paths intersect geometrically but the vehicles cannot plausibly occupy the shared area at overlapping times, the relationship does not need to impose the same immediate constraint.

Graph-based representations are appropriate for this type of multi-vehicle interaction because vehicles can be represented as nodes and directed conflict or precedence relations as edges [12]. The project uses this idea specifically for right-of-way relationships rather than treating the graph itself as a centralized crossing scheduler.

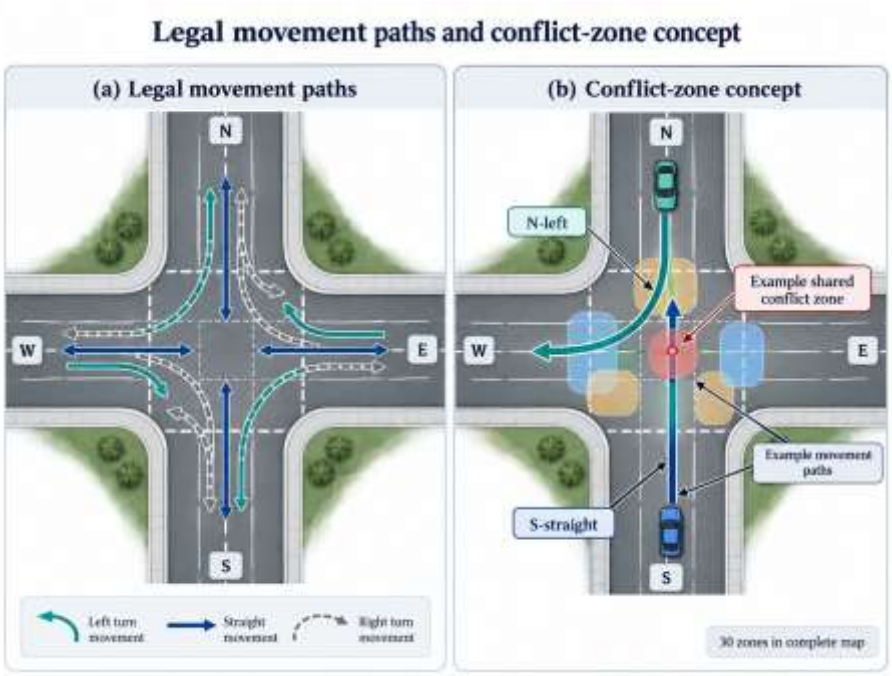


Figure 3 : Representation of legal intersection movements and map-derived conflict zones

4.4.4 Traffic-Rule Precedence and Claim Exchange

Once a physical and temporal conflict is identified, the traffic-rule layer determines whether one vehicle has a mandatory yield obligation.

The project uses the experimental profile DE_STVO_UNCONTROLLED_4WAY_V1, derived from a reviewed snapshot of the German Straßenverkehrs-Ordnung (StVO). The implemented profile includes equal-priority uncontrolled-intersection precedence under §8 and turning-related relationships under §9. A coordination-related interpretation of §11 is used only where explicit relinquishment is permitted by the project's protocol. The legal source is treated as an experimental regulatory profile rather than as a claim of legal certification.

A directed edge

A → B

means that **vehicle A yields to vehicle B**.

Each ego vehicle constructs precedence claims from its own current reasoning and publishes structured claim information. The communication baseline assumes an ideal same-step message exchange. It does not model a complete wireless V2V network and therefore does not claim realistic latency, packet loss, authentication, or communication security.

Mandatory obligations are protected from negotiation. Where the source evidence is stale, inconsistent, or insufficient, the system behaves conservatively instead of allowing MAPPO to decide on a potentially mandatory legal relationship.

4.4.5 Joint-Local Graph and MAPPO Negotiation

Received claims are combined with the ego vehicle's own claims to create a **joint-local precedence graph**. This graph belongs to the ego vehicle; it is not an omniscient global state shared directly by every actor.

The final neural representation uses 8-dimensional node features and 9-dimensional edge features. An edge-aware Message Passing Neural Network (MPNN) produces vehicle and graph representations using a hidden dimension of 64 and three message-

passing layers. Message-passing networks are designed to aggregate information through graph relationships and therefore provide a natural representation for directed multi-agent interactions [20].

In the final controlled MAPPO training configuration, the graph encoder remains fixed while the role-specific actor heads and centralized critic are trained. Two actor roles are used:

Proposer

- KEEP_CLAIM
- RELINQUISH_CLAIM

Responder

- ACCEPT_RELINQUISHMENT
- REJECT_RELINQUISHMENT

A relinquishment proposal alone does not change precedence. The corresponding counterparty must explicitly accept it before the deterministic protocol updates the effective coordination relationship.

The Boolean action mask is applied before sampling from the policy distribution. Therefore, a learned action can only be selected if it is valid for the current claim, role, and protocol state. This preserves the separation between learning which valid negotiation choice to make and deterministically deciding which choices are allowed.

4.4.6 Decentralized Physical Execution

Negotiation ends with semantic coordination commitments, not with physical vehicle commands. These commitments are passed to an execution controller owned by each ego vehicle.

The controller receives only information relevant to that ego, including its own state, commitments involving that vehicle, and fresh sender-owned physical-state information from relevant peers. It does not receive a global READY/BLOCKED plan or the centralized training critic.

Before an ego vehicle is admitted to a conflict zone, the controller checks whether each relevant physical relationship is ordered, cleared, or non-conflicting. An unresolved physical pair results in a safe hold rather than permission for both vehicles to proceed.

For negotiation-eligible relationships where a valid MAPPO outcome is rejected by the physical admissibility check, the repaired runtime can apply a deterministic local fallback using published queue-arrival information. This fallback is not allowed to override protected mandatory traffic-rule relationships and is recorded separately during evaluation rather than being reported as a direct MAPPO success.

Once valid admission has been established close enough to the conflict-zone boundary, the execution state prevents an impossible mid-zone reversal that would require a vehicle to stop after it has already committed to entering. Speed constraints are finally applied through TraCI while SUMO's native safety behavior remains active.

Dense traffic is handled without increasing the MAPPO graph to all live vehicles. Only the physical queue head on each incoming approach is eligible to negotiate, giving at most four simultaneous learned participants. Following vehicles remain under normal SUMO car-following until they become queue heads.

This design allows the simulation to contain more than four vehicles while preserving the agent-count assumptions under which the negotiation policy was developed.

4.5 Standards and professional practice

The project is a simulation-based research prototype and does not claim automotive safety certification. Nevertheless, several professional safety principles were considered during its design.

First, regulatory reasoning is based on a versioned snapshot of the German StVO rather than undocumented informal right-of-way assumptions. The project records the regulatory profile and source provenance so that rule interpretation can be audited independently of the learned policy.

Second, architecture follows a defense-in-depth approach. Learned decisions are limited by explicit traffic rules, geometric conflict reasoning, action masks, post-negotiation physical admissibility checks, ego-local execution constraints, and SUMO-native safety behavior. This is consistent with the wider automotive principle that safety should not depend on a single decision mechanism.

ISO 26262 provides the principal functional-safety framework for safety-related electrical and electronic road-vehicle systems, while ISO 21448:2022 (SOTIF) addresses unreasonable risk arising from functional or performance insufficiencies in intended functionality, particularly where situational awareness and complex processing are safety relevant. The present prototype was not developed through either certification lifecycle, so these standards are treated only as professional design context rather than standards with which the project claims compliance.

Professional practice was also supported through configuration control and reproducibility. Model checkpoints, rule profiles, scenario manifests, evaluation evidence, and policy hashes were retained as versioned research artefacts. Training operations were separated from evaluation, and the final validation path checks that actor parameters remain unchanged during inference.

Finally, the system clearly distinguishes simulation evidence from real-world safety evidence. SUMO experiments are used to study feasibility and limitations under controlled conditions; successful simulation outcomes are not presented as proof of public-road deployment readiness.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The right-of-way negotiation framework was implemented primarily in **Python** and integrated with **SUMO** through TraCI. The implementation was organized as a set of independent but connected modules for perception, intention prediction, conflict reasoning, traffic-rule evaluation, negotiation, execution, training, and evaluation. This modular structure was used so that each stage could be tested separately before being exercised within the complete runtime pipeline.

SUMO provided a microscopic traffic environment and vehicle lifecycle, while TraCI was used to read vehicle states and apply runtime speed constraints. This choice provided a repeatable environment in which the same intersection configuration and traffic schedules could be executed under controlled conditions [19].

The intention-prediction models were deployed in **ONNX** format and executed using **ONNX Runtime** on the CPU. This allowed the final SUMO runtime to perform intention inference without requiring the original TensorFlow training environment. The deployed bundle contains two GRU models for 1.0 s and 0.5 s prediction horizons, together with the feature specification, scaling parameters, calibration policy, and verification artefacts.

The MAPPO training and neural graph components were implemented using **PyTorch**. PyTorch was used only for the learning-related parts of the system; the deployed intention models remained independent ONNX artefacts. PyTorch is suitable for this type of research implementation because it supports flexible Python-based construction and training of neural models [24].

Additional libraries included NumPy for numerical operations, Shapely for geometric processing, and pytest for automated verification. A separate local presentation application was implemented using **NiceGUI**, providing read-only views of the live Component 2 simulation and recorded evaluation evidence.

Table 9 : Main implementation technologies

Technology	Use this research
Python	Core implementation and experiment orchestration
SUMO	Microscopic traffic simulation
TraCI / sumolib	Runtime simulation access and network interaction
PyTorch	MAPPO actors, critic, graph-learning and training infrastructure
ONNX Runtime	CPU inference for deployed GRU intention models
NumPy	Numerical and feature-processing operations
Shapely	Path and conflict-zone geometry
pytest	Unit, contract and integration testing
NiceGUI	Read-only project demonstration and evaluation interface

5.2 Component implementation

The implementation was organized according to the functional stages of the proposed architecture rather than as a single control program. Table 10 summarizes the major implementation areas.

Table 10 : Main implementation areas

Implementation area	Main responsibility
Ego-local observation	Vehicle observations and Local Dynamic Maps
Intention inference	ONNX-based maneuver prediction
Conflict reasoning	Movement paths, conflict zones and temporal occupancy
Regulatory reasoning	Mandatory right-of-way precedence
Negotiation learning	Claims, graph encoding, GNN and MAPPO actors
Policy training	PPO training, experiments and policy selection
Decentralized execution	Ego-local permissions and speed constraints
Runtime simulation	Continuous SUMO execution and telemetry
Targeted validation	Reproducible scenario-based validation
Post-training evaluation	Extended evaluation and baseline-comparison infrastructure

5.2.1 Perception and Intention Runtime

At every simulation step, relevant vehicle states are obtained from SUMO and converted into ego-local observations. Each autonomous vehicle maintains its own Local Dynamic Map containing the locally observed vehicles and their recent motion histories.

The intention-prediction stage transforms these histories into the same 48 × 6 causal feature representation used during model development. Two deployed GRU models estimate the probability of the LEFT, RIGHT, and STRAIGHT maneuver classes. The primary model provides a 1.0 s prediction horizon, while the secondary model operates at 0.5 s.

The deployment package retains the model input contract and preprocessing information used during development. This helps ensure that the feature representation used during SUMO execution is consistent with the representation on which the intention models were evaluated.

5.2.2 Conflict and Regulatory Reasoning

Predicted maneuvers are associated with the predefined intersection movement paths. The conflict-reasoning stage then determines whether candidate movements share a physical conflict region and whether the vehicles could occupy that region at overlapping times.

Regulatory reasoning is implemented independently from the learned negotiation mechanism. Directed precedence relationships are produced before MAPPO is considered, allowing mandatory traffic-rule decisions to remain inspectable and protected from learned modification.

This separation follows the design motivation identified in earlier right-of-way-based reinforcement-learning work, where explicit regulatory mechanisms were required to improve rule compliance [6].

5.2.3 MAPPO Negotiation Implementation

The learned stage operates on semantic right-of-way relationships rather than directly controlling vehicle acceleration or steering.

Local precedence claims received from relevant vehicles are combined with the ego vehicle's own claims to construct a joint-local interaction graph. An edge-aware message-passing representation is then used to encode the graph before the proposer and responder actors make their decisions. Message Passing Neural Networks are suited to relational problems in which information is propagated between connected entities [20].

The proposer selects between:

- KEEP_CLAIM
- RELINQUISH_CLAIM

The responder selects between:

- ACCEPT_RELINQUISHMENT
- REJECT_RELINQUISHMENT

State-dependent Boolean masks remove invalid actions before policy sampling. Therefore, MAPPO selects only among negotiation choices that are valid for the current protocol state. The general use of invalid-action masking in policy-gradient methods is discussed in [15].

The predefined E5, E10, and E15 configurations were compared during model selection. **E5**, which used five PPO update epochs, achieved the lowest mean validation Total Team Travel Time among the eligible tested candidates and was frozen for subsequent system evaluation.

5.2.4 Decentralized Physical Execution

The output of negotiation is a set of coordination commitments rather than a global driving command.

Each autonomous vehicle runs its own execution controller using the information relevant to that ego vehicle. The controller does not receive the centralized MAPPO training critic or a global intersection-wide GO/STOP decision.

Before entering a conflict zone, the ego controller verifies whether the relevant physical relationship has been resolved, cleared, or identified as non-conflicting. Missing, stale, contradictory, or unresolved information results in a conservative **safe-hold** decision.

When a valid negotiation result cannot be physically admitted, the implementation can use a deterministic local fallback based on the available queue-arrival information. This fallback remains subordinate to mandatory traffic-rule precedence and is recorded separately during evaluation.

For continuous traffic, only the current queue head on each approach can participate in learned negotiation. Therefore, the simulation may contain more than four active vehicles while the MAPPO negotiation context remains bounded to a maximum of four agents.

5.3 Integration with the other components

The broader project contains four research components under the common theme **Agentic AI for Robust and Socially-Aware Autonomous Driving**. The components share a common research objective but remain technically independent implementations.

Component 2 therefore does not directly depend on the pedestrian-intention model developed for Component 1 or on the internal models of Components 3 and 4. Its vehicle-intention model, right-of-way reasoning, MAPPO negotiation, and physical execution pipeline are self-contained.

Project-level integration was achieved through the common SOCIALMIND DRIVE showcase. This interface provides a shared entry point to the four research components while preserving their individual scientific evidence.

For the right-of-way component, the interface provides a component overview, live SUMO simulation, vehicle-level state inspection, quantitative evaluation results, and targeted validation evidence. The presentation layer reads recorded or live backend state but does not alter the scientific metrics or modify the selected policy.

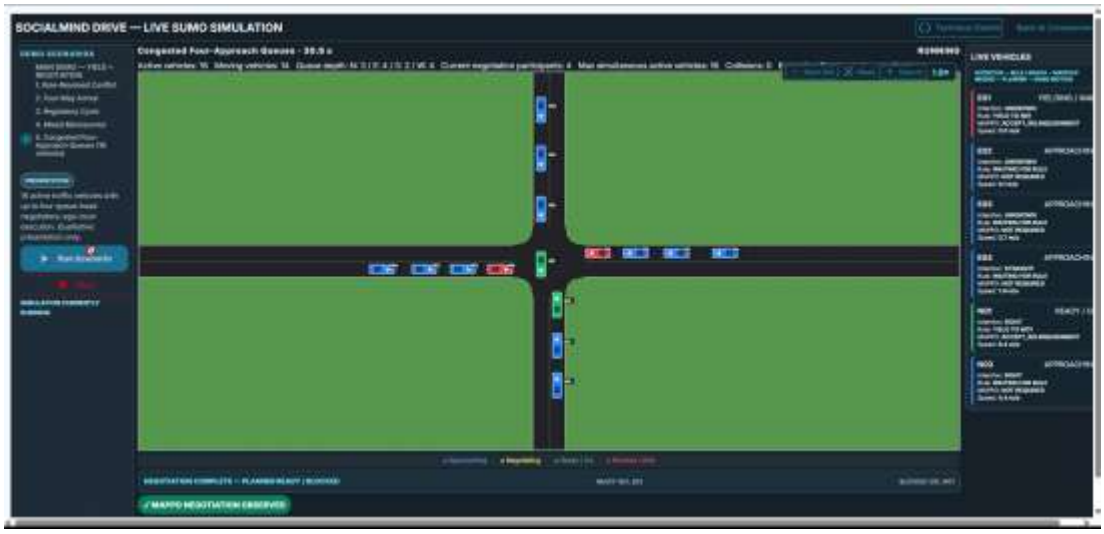


Figure 4 : Runtime simulation of continuous four-approach traffic

5.4 Verification and software testing

The implementation was verified at several levels. Unit and contract tests were used to check individual operations such as observation processing, feature construction, conflict reasoning, regulatory precedence, graph encoding, action masks, and negotiation transitions.

Integration tests were then used to verify the connections between the main processing stages, including the sequence from Local Dynamic Map construction to conflict assessment, traffic-rule reasoning, MAPPO negotiation, and decentralized execution.

Specific integrity tests were also implemented for the assumptions that support the decentralized research claim. These checks verify that:

- MAPPO cannot override protected mandatory precedence;
- the centralized critic is not called during deployed execution;
- no backward pass, optimizer step, or parameter update occurs during evaluation;
- the selected policy remains unchanged during inference;
- a global audit graph is not used as the physical execution authority;
- no more than four queue-head agents participate simultaneously in MAPPO; and
- Missing or stale coordination evidence produces safe holding rather than unsafe entry.

The project recorded a major Component 2 regression checkpoint with **704 passing tests** before later dense-traffic evaluation changes. Subsequent changes that primarily affected evaluation infrastructure were verified using focused test suites. The targeted-validation update recorded **39 focused passing tests**, while the later scalability-evaluation wrapper recorded **83 focused passing tests**. These focused checks are reported separately rather than implying that the complete regression suite was rerun after every evaluation-only modification.

Importantly, the testing process did not redefine unsuccessful traffic outcomes as successful simply because they remained collision-free. Liveness failures observed under difficult dense scenarios were retained as research evidence and are analyzed in Chapter 6.

Table 11 : Verification strategy

Test level	Representative checks
Unit	Features, graphs, rules and action masks
Contract	CTDE boundaries and policy semantics
Integration	Observation → negotiation → execution
Runtime simulation	Completion, zone entry and queue-head behavior
Fault testing	Missing and stale coordination information
Evidence integrity	Frozen configurations, policy hashes and recorded outputs

5.5 Deployment and operation

The completed component is operated as a **local simulation-based research prototype** rather than as an embedded autonomous-vehicle application.

The selected E5 policy can be loaded into the runtime environment for inference without performing additional training. A separate continuous runtime simulation supports higher-density four-approach traffic and produces telemetry for qualitative inspection.

Scientific validation is kept separate from the live demonstration. The **targeted validation suite** executes predefined rule-resolved, negotiation-required, complex, congested, and fault-oriented scenarios and records immutable evidence for subsequent analysis.

A separate **post-training evaluation framework** supports a larger set of scenarios, comparison methods, metrics, replay functions, provenance checks, and report generation. This infrastructure is intentionally separated from model training so that evaluation cannot silently update the selected policy.

The common project showcase provides a read-only presentation layer over the simulation and stored evaluation evidence. It therefore supports demonstration and inspection without becoming part of the right-of-way decision process.

The deployment should be interpreted as a reproducible research environment. It provides simulation, model inference, experiment replay, evidence recording, and presentation functions, but it is not a production V2V communication system or a road-certified autonomous-driving deployment.

6 RESULTS AND VALIDATION

6.1 Research results

The results are presented in three stages: performance of the intention-prediction models, selection of the MAPPO negotiation policy, and system-level validation of the completed right-of-way framework. The results are reported within the bounded simulation scope defined in Chapter 3 and are not interpreted as evidence of real-world deployment safety.

6.1.1 Intention-Prediction Results

The two deployed GRU intention models were evaluated on the held-out inD test recording. Both models classified vehicle movements into **left, right, and straight** manoeuvres using the causal motion-history features described in Chapter 3.

Table 12 : Test performance of the deployed intention models

Model	Lead time	Accuracy	Balanced accuracy	Macro F1	Log loss
Primary GRU	1.0 s	89.66%	89.10%	88.93%	0.425
Secondary GRU	0.5 s	93.10%	91.67%	92.38%	0.245

Both models were evaluated on 58 test samples. The 0.5 s model achieved the stronger overall performance, while the 1.0 s model provided earlier prediction at a modest reduction in classification performance. The primary model achieved class-level F1-scores of 0.903 for left turns, 0.824 for right turns, and 0.941 for straight movements. These results support the use of the models as intention evidence within the subsequent conflict-reasoning pipeline.

The inD dataset is used here only for the intention-estimation component; the complete right-of-way framework remains validated in SUMO rather than on real-road vehicle interactions, consistent with the scope stated earlier. The dataset itself was developed specifically from naturalistic intersection trajectories [8].

6.1.2 MAPPO Policy Selection

The final MAPPO selection compared three predeclared configurations that differed only in the number of PPO update epochs. All three candidates satisfied the recorded collision and blocked-zone safety gates during validation.

Table 13 : MAPPO candidate selection results

Candidate	PPO epochs	Mean validation TTT (s)	Median TTT (s)	SD (s)	Collisions / blocked violations
E5	**5**	**15,817.91**	15,499.12	4,909.04	0 / 0
E10	10	16,579.07	16,385.40	3,112.97	0 / 0
E15	15	16,942.56	19,127.88	4,325.35	0 / 0

Because lower Total Team Travel Time (TTT) was the predefined selection objective, **E5 was selected and frozen** for subsequent evaluation. The result should be interpreted only as the best of the three tested configurations; it does not establish that five epochs are globally optimal.

After selection, E5 was evaluated on the held-out scenario split across 108 scenario executions. The recorded mean TTT was 426.64 s, with zero collisions and zero blocked-zone violations across those executions. Held-out data was not used to select E5.

A complete efficiency comparison against the deterministic no-negotiation baseline was not considered valid because that baseline failed the predefined hard safety gate in one held-out scenario. Therefore, this thesis should not claim that MAPPO outperformed the baseline overall.

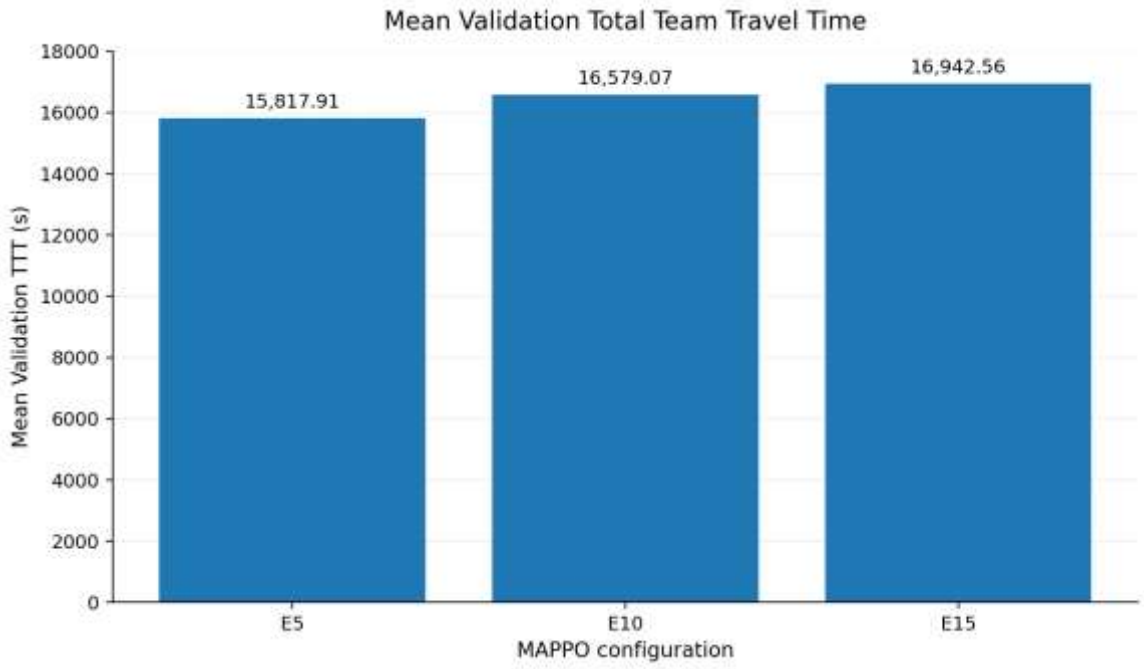


Figure 5: Mean validation Total Team Travel Time for the predefined MAPPO candidate configurations.

Data used: E5 = 15,817.91 s; E10 = 16,579.07 s; E15 = 16,942.56 s.

6.1.3 Targeted System Validation

The selected policy was subsequently evaluated through a compact targeted-validation suite. Only the current V2 evidence is summarized in Table 14, preserved historical duplicates are not repeated in the thesis.

Table 14 : Summary of targeted system-validation results

Case	Vehicles	Completed	Mean wait (s)	MAPPO calls	Safety	Result
PV-01 Rule-resolved	2	2/2	0.50	0	0 / 0	PASS
PV-02 Negotiation cycle	4	0/4	15.11	0	0 / 0	SAFE_DEGRADATION
PV-02 Rules-only	4	0/4	15.11	0	0 / 0	SAFE_DEGRADATION
PV-03 Multiple left turns	3	3/3	4.24	1	0 / 0	PASS
PV-04A Nominal dense	16	1/16	141.87	2	0 / 0	SAFE_DEGRADATION
PV-04B Critical dense	16	0/16	35.79	0	0 / 0	SAFE_DEGRADATION
PV-05 Four-way simultaneous	4	0/4	15.89	0	0 / 0	SAFE_DEGRADATION
PV-06 Message fault checks	–	–	–	–	Fail-safe pass	PASS

The results demonstrate two different behaviors. PV-01 confirmed correct rule-resolved operation without requiring MAPPO. PV-03 provided the clearest successful learned-negotiation case: all three vehicles completed the scenario without a collision or blocked-zone violation. MAPPO was invoked once, although the resulting event required a deterministic safety fallback before physical execution.

The difficult cases exposed a different limitation. PV-02 and PV-05 remained physically safe but did not achieve traffic completion. These outcomes were therefore retained as SAFE_DEGRADATION rather than being classified as successful.

The distinction between safety and liveness is important. A collision-free episode is not sufficient evidence of effective intersection management when the vehicles remain blocked.

6.1.4 Dense-Traffic and Fairness Results

PV-04A directly evaluated the congested scenario requested during project review. The case scheduled 16 vehicles, four from each approach, while restricting the learned negotiation context to at most four physical queue-head vehicles.

All 16 vehicles were successfully spawned, and the episode reached **16 simultaneously active and 16 simultaneously moving vehicles**. However, only **1 of the 16 vehicles completed** within the evaluation period. No collision or blocked-zone violation occurred, giving a completion rate of **6.25%** and throughput of **0.333 vehicles/minute**.

The waiting-time results were:

North = 116.59 s

East = 150.95 s

South = 153.74 s

West = 146.20 s

The resulting approach waiting-time gap was **37.15 s**. Fairness is reported separately because previous autonomous-intersection research has shown that an efficiency-focused result can hide unequal treatment between vehicles or approaches [13].

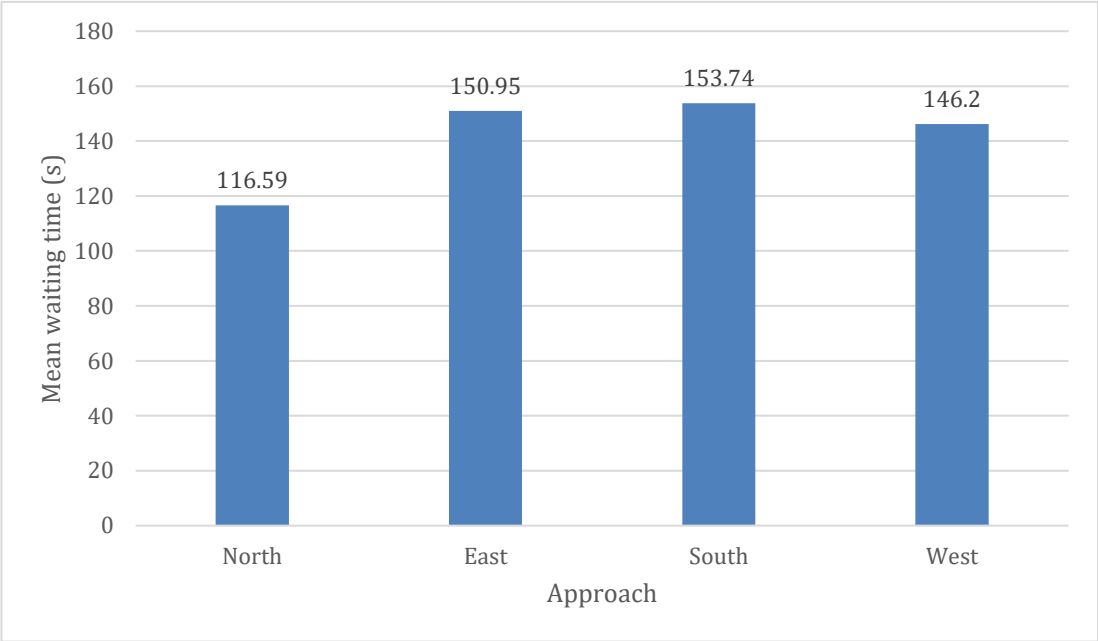


Figure 6 : Mean waiting time by approach in the PV-04A nominal dense validation case

The fairness metrics were successfully measured, but a strong fairness conclusion cannot be drawn from PV-04A because the episode experienced severe liveness degradation.

6.2 Product and technical results

Beyond the scenario-level results, the implementation was verified against the architectural requirements described in Chapters 4 and 5.

The targeted-validation records confirmed that runtime execution remained decentralized. Across the recorded validation evidence, the centralized MAPPO critic was not called during runtime, the global audit graph was not used as the physical execution authority, no training or optimizer operations occurred, and the selected E5 policy hash remained unchanged.

The latest naming-only repository refactor was also verified without rerunning scientific experiments. **285 focused tests passed**, while all recorded historical scientific evidence and the selected E5 checkpoint remained unchanged. This confirms that the professional renaming of the active runtime, targeted-validation, and post-training-evaluation packages did not alter the underlying scientific results.

A separate continuous runtime demonstration was also used to verify the presentation behavior requested by the panel. In this **qualitative presentation run**, 16 vehicles were scheduled, spawned, and completed in **82.76 simulated seconds**, with a maximum of 16 active vehicles and no collision or blocked-zone entry violation. The learned negotiation set remained bounded to four queue-head vehicles.

One SUMO emergency-braking warning was retained in that run rather than being hidden from the evidence.

This qualitative result should not replace PV-04A. The two runs serve different purposes:

- The **demonstration** shows that the implemented system can visibly process continuous four-approach traffic under the presentation schedule;
- The **targeted validation case** provides the frozen scientific evidence and exposes the dense-traffic liveness limitation.

Keeping these two forms of evidence separate prevents a successful demonstration from being used to conceal a more difficult validation result.

6.3 User or stakeholder validation

No public-road user study or human-subject usability experiment was performed. Stakeholder validation in this project was instead based on the technical feedback received during research reviews and on whether the final prototype provided measurable evidence addressing those concerns.

Table 15 : Response to project-review feedback

Review concern	Validation response	Observed evidence
Fairness and time measurement should be included	Waiting, throughput and per-approach fairness metrics added	PV-04A reported mean/max wait and 37.15 s approach gap
Complex scenarios should be demonstrated	Multiple-left-turn and simultaneous four-way cases included	PV-03 PASS; PV-05 exposed a liveness limitation
Four vehicles were insufficient for congestion testing	Continuous four-approach dense traffic added	PV-04A/B used 16 vehicles; max 16 active vehicles observed
System should remain decentralized	Runtime integrity checks added	Critic runtime calls = 0; global execution planner not used

This evidence shows that the requested evaluation areas were implemented and measured. It does not mean that every requested scenario produced a successful outcome. In particular, the congestion-related feedback led directly to the identification of a liveness limitation that would not have been visible in the earlier four-vehicle demonstrations.

6.4 Integration results

At project level, the right-of-way component was integrated into the **SOCIALMIND DRIVE** presentation environment together with the other three research components. The Component 2 interface displays the live SUMO simulation, current vehicle state, right-of-way information, and recorded validation evidence.

The integration remained at presentation-level rather than algorithmic. Component 2 continued to use its own vehicle-intention models and right-of-way pipeline, while the other project components retained their own models and experimental workflows.

The integration therefore demonstrates that the four research components can be presented within one coherent autonomous-driving project without changing the scientific boundary of the individual thesis.

The runtime checks also supported the Component 2 decentralization claim: no centralized critic was used for action selection, no global execution graph issued vehicle commands, and the learned negotiation group remained bound to the physical queue heads. MAPPO itself follows the broader centralized-training/decentralized-execution approach described by Yu et al., in which actors use their observations during execution while centralized information can assist critic training.

6.5 Summary of findings

The experimental results provide a mixed but informative outcome.

The intention-prediction component achieved approximately **89.7% accuracy at a 1.0 s lead time** and **93.1% accuracy at a 0.5 s lead time**. Among the three tested MAPPO configurations, E5 achieved the lowest mean validation TTT and was selected without using held-out scenarios for model selection.

At system level, the framework successfully completed the rule-resolved PV-01 case and the complex three-vehicle PV-03 case without collision or blocked-zone violations. The dense 16-vehicle validation demonstrated that the implementation could maintain a bounded four-agent negotiation context while substantially more traffic was active.

However, the dense validation did not establish successful traffic discharge. PV-04A completed only one of sixteen vehicles, while PV-04B completed none. Several smaller critical scenarios also resulted in safe degradation. The principal system-level limitation observed in the final validation is therefore **liveness under difficult, decentralized precedence conditions**, rather than an observed collision failure.

The results support a bounded conclusion: the proposed hybrid architecture can perform correct rule resolution and learned negotiation in selected intersection interactions, while its current conservative reconciliation mechanism requires further improvement before it can reliably handle dense continuous traffic.

7 DISCUSSION

7.1 Interpretation of findings

The results show that the proposed framework performs differently depending on the structure of the intersection interaction. In simple rule-resolved conditions, the deterministic reasoning layer was sufficient to establish a valid crossing order without requiring learned negotiation. PV-01 therefore demonstrates that architecture can avoid unnecessary use of MAPPO when traffic-rule precedence already provides a clear result.

The more complex PV-03 case provides evidence that the learned negotiation pathway is operational. All three vehicles completed the multiple-left-turn scenario, and MAPPO was invoked during the coordination process. However, the MAPPO decision required a deterministic safety fallback before physical execution. Therefore, PV-03 should not be interpreted as evidence that the learned policy alone resolved the complete scenario. Instead, it demonstrates the intended hybrid behavior of the system, where learned negotiation operates within deterministic safety constraints.

Different behavior was observed in the dense and simultaneous cases. PV-04A scheduled 16 vehicles and reached 16 simultaneously active vehicles, but only one vehicle completed within the evaluation budget. PV-04B and several smaller critical cases also remained incomplete. Importantly, these runs recorded no collisions or blocked-zone violations. The dominant limitation was therefore **liveness rather than observed physical safety**.

The recorded traces provide a technical explanation for this behavior. Different ego vehicles can infer different maneuver intentions for the same neighboring vehicle. These differences can create reciprocal mandatory precedence claims in the local graphs. Because the implementation deliberately prevents MAPPO from overriding relationships that may represent mandatory traffic obligations, such disagreement results in conservative holding rather than forced negotiation. This behavior is consistent with the safety-first design, but under dense traffic it can prevent progress.

The fairness measurements should be interpreted with similar caution. PV-04A produced measurable differences between the four approaches, including a 37.15 s approach waiting-time gap. However, because only one of the sixteen vehicles completed, this case cannot establish that the system provides fair traffic service. The measurements are more useful as evidence that fairness is observable and can be analyzed independently from collision safety.

7.2 Comparison with literature baselines and existing solutions

The findings are consistent with earlier research showing that autonomous intersection coordination requires more than a single decision mechanism. DCL-AIM demonstrated that decentralized multi-agent learning can dynamically identify coordination needs rather than treating every vehicle as part of one large decision problem [5]. The present framework follows a related principle, but identifies relevant interactions through movement geometry, temporal conflict assessment, and queue-head eligibility before learned negotiation is considered.

Zhuang et al. proposed a decentralized attention-based MAPPO framework in which neighboring information, traffic rules, and trajectory prediction contribute to cooperative decision-making [11]. In contrast, the present work uses deterministic conflict and regulatory reasoning to reduce the decision space before MAPPO. This produces a more explicit boundary between mandatory precedence and learned negotiation, although the current results show that inconsistent local beliefs can still prevent liveness.

The results also support the motivation identified by Feng et al., who reported that reinforcement learning alone did not guarantee full right-of-way compliance and therefore introduced a dedicated right-of-way cooperator [6]. Their approach operates within a centralized vehicle-road cooperative architecture, while the present research attempts to retain ego-local runtime execution. The comparison therefore supports the use of explicit right-of-way constraints but does not establish that the proposed decentralized approach performs better than their system.

Fairness research in Autonomous Intersection Management has also shown that optimization for efficiency alone can disadvantage individual vehicles or approaches [13]. For this reason, the present study reports approach-specific waiting times and service-order information rather than relying only on average travel time or throughput.

A complete quantitative superiority claim against FCFS, rules-only, or other MARL methods is not supported by the current evidence. The deterministic held-out baseline failed a predefined safety-validity gate, while several targeted MAPPO

cases experienced liveness degradation. The contribution of the thesis should therefore be interpreted primarily as an architectural and experimental contribution rather than proof of overall performance superiority.

7.3 Research contribution

The main research contribution is a **hybrid decentralized right-of-way architecture** that separates mandatory reasoning from learned negotiation.

The framework combines ego-local observations, learned movement-intention estimation, map-based conflict geometry, temporal conflict relevance, explicit regulatory precedence, graph-structured claim exchange, constrained MAPPO negotiation, and ego-owned physical execution.

A particularly important aspect is the restriction placed on the learned policy. MAPPO does not directly control steering or acceleration and cannot intentionally remove a protected mandatory precedence relationship. Instead, it operates on a small semantic negotiation vocabulary only when the preceding reasoning layers identify a relationship as negotiable.

The research also contributes evidence about where this architecture becomes difficult to operate. The dense scenarios show that decentralized safety-oriented reasoning can produce conflicting local precedence beliefs and conservative deadlock. Identifying this limitation is itself relevant because it exposes a problem that may be hidden in simpler four-vehicle demonstrations.

7.4 Product contribution

The product contribution is a modular research prototype that connects intention prediction, right-of-way reasoning, learned negotiation, decentralized execution, SUMO simulation, evidence recording, and scenario replay within one implementation.

The prototype also includes reproducible targeted validation, runtime integrity checks, and a read-only presentation interface. These facilities make it possible to inspect not only a final movement decision but also the conditions under which MAPPO was invoked, whether a deterministic fallback was required, and whether the deployed execution remained decentralized.

The resulting system should be viewed as a **research and validation prototype**, not as a production autonomous-driving controller.

7.5 Limitations and threats to validity

The study has several limitations that affect how the results should be interpreted.

The system-level experiments use one unsignalized four-way SUMO intersection and autonomous vehicles only. Human-driven vehicles, pedestrians, cyclists, different intersection geometries, poor road conditions, and real sensor uncertainty are not included. The results therefore have limited environmental generalizability.

The communication model is also simplified. Coordination messages are exchanged within simulation without modelling a complete V2V communication stack, realistic network delay, packet loss, malicious messages, or authentication.

The intention models were developed using a bounded subset of the inD dataset. Although real-world trajectory data improves the realism of intention modelling, the final right-of-way framework was not validated on physical vehicles.

A further limitation is the maximum learned negotiation group of four queue-head agents. This design allows more traffic to exist in the simulation but does not demonstrate that the learned negotiation policy itself scales to dozens of simultaneous agents.

The separate 4–40 vehicle scalability study was prepared but not completed. Its V2 preflight experienced safe deadlock with four vehicles, and no scientific seven-load series was recorded. Therefore, the thesis should not claim a measured scalability threshold.

Finally, the dense-traffic results show a clear liveness limitation caused by inconsistent ego-local intention and precedence beliefs. Solving this problem would require an additional decentralized reconciliation mechanism rather than simply allowing MAPPO to override possibly mandatory traffic obligations.

7.6 Implications

The main implication of the study is that decentralization and safety constraints alone do not guarantee effective intersection flow. Conservative architecture can remain physically safe while still failing to make progress.

This suggests that future decentralized intersection systems need mechanisms not only for sharing decisions but also for reconciling the **premises** from which those decisions were derived, particularly uncertain maneuver intentions and regulatory interpretations.

The results also support the value of reporting safety and liveness independently. A collision-free system should not automatically be described as successful when traffic cannot be discharged.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis investigated **Multi-Agent Negotiation for Right-of-Way in Complex Intersections** with the aim of developing a decentralized framework for autonomous vehicles operating at an unsignalized four-way intersection.

The final architecture combines ego-local observation, GRU-based movement-intention estimation, spatial and temporal conflict reasoning, explicit traffic-rule precedence, constrained MAPPO negotiation, and decentralized physical execution.

The experimental results show that the framework can correctly resolve selected interactions. The rule-resolved scenario was completed without requiring MAPPO, while the three-vehicle multiple-left-turn scenario successfully completed using the learned negotiation pathway together with the deterministic safety fallback. Across the targeted validation evidence, no collisions or blocked-zone violations were observed and runtime integrity checks confirmed that execution remained decentralized.

The results also exposed an important limitation. Under dense and simultaneous traffic, different vehicles could derive conflicting mandatory precedence claims from inconsistent local intention beliefs. Because the architecture conservatively protects potentially mandatory obligations, these disagreements could lead to safe holding and deadlock. The current prototype therefore demonstrates **bounded simulation-level feasibility rather than complete dense-traffic capability**.

The study concludes that constrained learned negotiation can form a useful part of a decentralized right-of-way architecture, but reliable operation under complex continuous traffic requires stronger decentralized reconciliation of uncertain intentions and precedence claims.

8.2 Achievement of objectives

Table 16 : Achievement of the research objectives

Objective	Evidence	Outcome
O1 – Ego-local traffic representation	Per-vehicle LDM and local runtime reasoning	Achieved
O2 – Movement-intention estimation	Two deployed GRU models using inD trajectories	Achieved
O3 – Conflict and regulatory reasoning	12 paths, 30 conflict zones and mandatory precedence layer	Achieved
O4 – MAPPO negotiation	Trained E5 semantic proposer/responder policy	Achieved within bounded scenarios
O5 – Decentralized physical execution	Critic runtime calls = 0 and no global execution authority	Achieved
O6 – System evaluation	Rule, complex, dense, fairness and fault-oriented validation	Achieved; dense liveness limitation identified

The objectives concerned the **development and evaluation** of the framework rather than a requirement for every scenario to pass. Therefore, the dense failures do not invalidate O6; instead, they provide evidence about the current operating limitation.

8.3 Individual research and product contributions

The individual research contribution is the design and evaluation of a hybrid decentralized right-of-way framework in which deterministic reasoning establishes non-negotiable precedence and MAPPO is limited to relationships that remain eligible for negotiation.

The product contribution is the implemented SUMO-based research prototype, including local perception and intention inference, conflict and regulatory reasoning, graph-based learned negotiation, ego-local physical execution, automated verification, reproducible evaluation, and replay/presentation support.

A further contribution is the explicit documentation of failure behavior. The final system does not conceal dense-traffic deadlock behind collision-free statistics; instead, safety, liveness, fairness, and efficiency are reported separately.

8.4 Future directions

Future work should first address the identified **decentralized claim-reconciliation problem**. Vehicles require a method for resolving conflicting intention or regulatory premises without giving a learned policy unrestricted authority over mandatory traffic rules.

A second direction is broader scalability evaluation. After the liveness problem is addressed, the same framework should be evaluated across systematically increasing traffic loads, additional seeds, and different intersection layouts.

Communication realism should also be increased by modelling latency, packet loss, message authentication, and potentially malicious or inconsistent V2V information.

The operational domain should then be expanded beyond the current AV-only four-way intersection to include mixed human-driven traffic, pedestrians, cyclists, different road-priority structures, roundabouts, and multi-intersection networks.

Finally, stronger validation should include hardware-in-the-loop, vehicle-in-the-loop, and eventually controlled physical-vehicle experiments. Such extensions would also require formal safety engineering aligned with relevant automotive functional-safety and SOTIF practices.

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Chapter 2 reference additions

Chapter 1 එකේ [1]–[8] continue කරලා මේවා add කරන්න.

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[Guidance - delete before submission: Include every source cited in the thesis and no uncited sources. Use one style consistently; IEEE is recommended for IT4010 unless the department instructs otherwise. Single-space each entry and leave one blank line between entries.]

[[1] Replace with the first complete reference entry.]

GLOSSARY

Term	Definition
Blocked-zone violation	A condition in which a vehicle enters or occupies a conflict region without satisfying the required physical admission conditions.
Conflict zone	A mapped region of the intersection where two or more legal vehicle movement paths cross or merge.
Ego vehicle	The autonomous vehicle from whose local observation and decision-making perspective the current situation is evaluated.
Ego-local observation	Information is available to an individual vehicle through its own simulated perception rather than through unrestricted access to the complete global simulation state.
Joint-local graph	A graph constructed by an ego vehicle using its own precedence claims together with relevant claims received from neighboring vehicles.
Liveness	The ability of the traffic system to continue making progress and allow vehicles to complete their routes rather than remaining indefinitely blocked.
Mandatory precedence	A right-of-way relationship established by the regulatory reasoning layer that cannot be removed by learned negotiation.
Negotiation-eligible relationship	A vehicle interaction for which mandatory rules do not fully determine the ordering and learned negotiation is therefore permitted.
Queue head	The leading vehicle on an incoming approach that is currently eligible to participate in intersection negotiation.
Safe degradation	An evaluation outcome in which no defined unsafe event is observed, but the required traffic demand is not successfully

	completed.
Safe hold	A conservative execution decision that prevents a vehicle from entering a conflict zone when coordination information is missing, stale, inconsistent, or unresolved.
Semantic negotiation action	A learned action that changes a right-of-way claim, such as keeping, relinquishing, accepting, or rejecting a claim, rather than directly controlling acceleration or steering.
Targeted validation	A predefined set of representative scenarios used to evaluate safety, liveness, negotiation behavior, congestion, fairness, and failure conditions.

APPENDICES

Appendix A Research instruments or extended results

This appendix provides supplementary validation and software-verification evidence supporting the results presented in Chapters 5 and 6. The information is included separately to keep the main thesis focused on the principal findings while retaining additional evidence relevant to reproducibility and research integrity.

A.1 Dense-Traffic Validation Evidence

Metric	**PV-04A nominal dense case**
Scheduled vehicles	16
Spawned vehicles	16
Maximum simultaneously active vehicles	16
Maximum simultaneously moving vehicles	16
Completed vehicles	1 / 16
Completion rate	6.25%
Mean waiting time	141.87 s
Throughput	0.333 vehicles/min
MAPPO calls	2
Collisions	0
Blocked-zone violations	0
Approach waiting-time gap	37.15 s
Final classification	SAFE_DEGRADATION

A.2 Per-Approach Waiting-Time Evidence

Approach	Mean waiting time
North	116.59 s
East	150.95 s
South	153.74 s
West	146.20 s

These values are provided as descriptive fairness evidence. They should not be interpreted as proof that the system achieved fair traffic service because the corresponding dense episode experienced severe liveness degradation.

A.3 Software Verification Checkpoints

Verification stage	Recorded result
Major Component 2 regression checkpoint	704 tests passed
Targeted-validation focused verification	39 tests passed
Scalability-evaluation focused verification	83 tests passed
Naming-only repository refactor verification	285 focused tests passed
Runtime centralized critic use	None observed during deployed evaluation
Policy modification during evaluation	None recorded

The verification checkpoints above were performed at different stages of development. They are therefore reported separately and should not be interpreted as

evidence that the complete regression suite was rerun after every evaluation-only modification.

Appendix B Academic integrity and AI use disclosure

Generative AI and automated coding tools were used as supporting tools during the development and documentation of this research. These tools were not treated as sources of experimental evidence, and the author remained responsible for the implementation, verification, interpretation, citations, and conclusions presented in the thesis.

Tool	Purpose in this research	Verification performed
ChatGPT (OpenAI)	Used to support technical explanation, thesis structuring, language refinement, literature organization, review of research presentation, and preparation of development prompts.	Generated text was reviewed against the implemented project, recorded experiment outputs, thesis requirements, and cited sources before inclusion.
OpenAI Codex	Used to assist with code organization, testing and evaluation infrastructure, repository refactoring, documentation, and reproducibility-related development tasks.	Code changes were reviewed using automated tests, recorded outputs, policy-integrity checks, and file-change inspection. Scientific results were not accepted solely from generated code descriptions.

Representative workflows included reviewing technical sections against the implementation, preparing prompts for evaluation and validation tooling, refining explanations of the decentralized architecture, and improving the organization of thesis material. Experimental metrics reported in the thesis were taken from executed project artefacts and retained validation evidence rather than generated by the language models. Long-running experiments that were not completed are explicitly identified as incomplete in the thesis.